



Fire Statement Fire Strategy Document

At

Belmont and Sutton Muslim Assembly (BASMA)
Former Ambulance Station,
Dorset Road,
Belmont, Sutton
SM2 6JH

Strategy Report

Date: 26th November 2025

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Validation and Compliance

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Report Issued	26th November 2025

Progress Register

Initial Fire Strategy Report and subsequent review dates	Status
30 th October 2025	Draft Fire Strategy Statement Report
26 th November 2025	Fire Strategy Statement Report

Author of this report.

The author of this report served over 30 years in the local authority's fire service in Kent. Promoted within the organisation via the fire services' statutory national examinations to a level of senior officer, also gaining reciprocal qualifications within the Institute of Fire Engineers [ife]. He attended all required courses for senior officers at the Fire Services' College Morton-in-Marsh Gloucestershire gaining further qualifications and experience. He has also attained the qualification of Level 4 Diploma for fire risk assessments through the Fire Protection Association with a third-party accreditation via ABBE and qualified as a fire door inspector via the UK Training Company and appears on the national data base as a qualified fire door inspector. He is also a member of the Institute of Fire Safety Management [MIFSM].

During his career he served in the various disciplines of both operational and fire safety disciplines.

On retirement from the fire service, he joined ATC Fire Safety as a consultant and remains to this day, which has extended his career by a further 18 years.

ATC Fire Safety Consultancy is an associate member of the ife [Institution of Fire Engineers] and FPA [Fire Prevention Association] and affiliated member of The Institute of Fire Safety Managers [IFSM] which allows all its members to keep up to date with all relevant fire safety matters.

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1.0 Executive Summary



This fire strategy report relates to a project for the design, redevelopment, and delivery of a community centre with prayer facilities at Belmont and Sutton Muslim Assembly (BASMA), Former Ambulance Station, Dorset Road, Belmont, Sutton SM2 6JH

The building is a former ambulance station which is located within Dorset Road, Sutton. Being a former Ambulance Station there is compliant access to the site when considering B5 of Approved Document B Volume 2 [Buildings other than dwellings], which considers Fire Appliance access.

The proposed use for the building is as a community centre with prayer facilities for the local Muslim community of Sutton.

The proposals are to retain the existing footprint of the building and extend, at ground floor level, to the south and west elevations of the building with the extension to the south elevation by approximately 4.3m and the west elevation by 16m. It is also proposed to raise the roof level above the men's prayer hall to form an atrium for visual effect allowing addition natural light into the building.

On this basis and to ensure compliance with various areas of the building, Approved Document B Volume 2 [Buildings other than Dwellings] will be used. This document will also refer to BR187 - External fire spread: building separation and boundary distances, and PAS:9980 Fire Risk Appraisal of External Walls and Cladding of buildings.

The property will be built of traditional building materials which will be compliant with Approved Document B Volume 2, [Table B2], with a reinforced concrete floor slab.

The internal walls will be either constructed of block/brick or studwork partitioning. Internal passive fire sub-compartmentation construction will offer a minimum of 30-minutes fire rating, and where required, these design parameters are increased to 60-minutes fire rating to remain compliant with building regulations.

The ground floor has an adequate provision of fire exits which also supports the principles of the angle of divergence where occupants can turn their backs on any seat of fire by more than 45° and move to an alternative escape route.

Travel distances for both single and alternative means of escape principles will be applied according to use and compliant with current building regulations.

The external wall system [EWS] will comply with building regulations for buildings under 18m in height to a minimum standard which requires non-combustible cladding of a **C-s3, d2** or better.

To clarify the above, **C** indicates the material has a limited contribution to a fire, **s3** refers to smoke development where it produces a substantial amount of smoke and **d2** refers to droplets produced when burning, which is considered a large amount in the first 10 minutes of exposure.

Table 12.1 Reaction to fire performance of external surface of walls			
Building type	Building height	Less than 1000mm from the relevant boundary	1000mm or more from the relevant boundary
'Relevant buildings' as defined in regulation 7(4) (see paragraph 12.15)		Class A2-s1, d0 ⁽¹⁾ or better	Class A2-s1, d0 ⁽¹⁾ or better
All 'residential' purpose groups (purpose groups 1 and 2)	More than 11m	Class A2-s1, d0 ⁽²⁾ or better	Class A2-s1, d0 ⁽²⁾ or better
	11m or less	Class B-s3, d2 ⁽²⁾ or better	No provisions
Assembly and recreation	More than 18m	Class B-s3, d2 ⁽²⁾ or better	From ground level to 18m: class C-s3, d2 ⁽³⁾ or better From 18m in height and above: class B-s3, d2 ⁽²⁾ or better
	18m or less	Class B-s3, d2 ⁽²⁾ or better	Up to 10m above ground level: class C-s3, d2 ⁽³⁾ or better Up to 10m above a roof or any part of the building to which the public have access: class C-s3, d2 ⁽³⁾ or better ⁽⁴⁾ From 10m in height and above: no minimum performance
Any other building	More than 18m	Class B-s3, d2 ⁽²⁾ or better	From ground level to 18m: class C-s3, d2 ⁽³⁾ or better From 18m in height and above: class B-s3, d2 ⁽²⁾ or better
	18m or less	Class B-s3, d2 ⁽²⁾ or better	No provisions

The roof is of a flat roof design.

The overall footprint for the building at ground floor level is 148.8m² approx.

The building is of ground floor accommodation only, therefore, the calculations regarding height of the building will be based on buildings below 5m. On this basis PAS 9980 principles will be adopted as there is no requirement for an EWS1 form to be completed as the building is below the required height and the risk of external fire spread is considered low. Also, adjacency to other buildings has been considered.

This fire strategy statement details the actions to be taken to ensure the building becomes both safe and compliant regarding fire safety matters.

This fire strategy document recognises this approach with the assumption that all permissions for planning and approval for the building have been gained up to this stage of the development.

The proposals outlined in this document demonstrate a level of fire safety equal to, or greater than the general standard implied by compliance with the recommendations in Approved Document B.

The occupation of the proposed project is detailed in Approved Document B, Volume 2 - Buildings other than Dwellings, Purpose Group 5 – Assembly and Recreation.

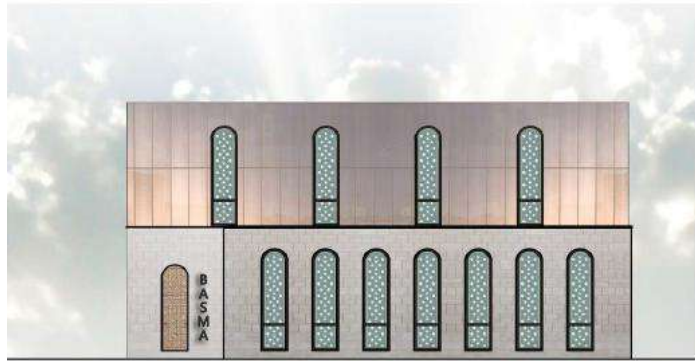
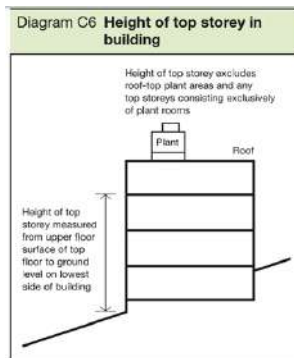
Table 0.1 Classification of Purpose Groups

Table 0.1 Continued		
Title	Group	Purpose for which the building or compartment of a building is intended to be used
Office	3	Offices or premises used for any of the following and their control: • administration • clerical work (including writing, bookkeeping, sorting papers, filing, typing, duplicating, machine calculating, drawing and the editorial preparation of matter for publication, police and fire and rescue service work) • handling money (including banking and building society work) • communications (including postal, telegraph and radio communications) • radio, television, film, audio or video recording • performance (premises not open to the public).
Shop and commercial	4	Shops or premises used for either of the following. • A retail trade or business (including selling food or drink to the public for immediate consumption, retail by auction, self-selection and over-the-counter wholesale trading, the business of lending books or periodicals for gain, the business of a barber or hairdresser, and the rental of storage space to the public). • Premises to which the public are invited either: – to deliver or collect goods in connection with their hire, repair or other treatment – (except in the case of repair of motor vehicles) where the public themselves may carry out such repairs or other treatments.
Assembly and recreation	5	Place of assembly, entertainment or recreation, including any of the following: • bingo halls, broadcasting, recording and film studios open to the public, casinos, dance halls • entertainment, conference, exhibition and leisure centres • funfairs and amusement arcades • museums and art galleries, non-residential clubs, theatres, cinemas, concert halls • educational establishments, dancing schools, gymnasia, swimming pool buildings, riding schools, skating rinks, sports pavilions, sports stadia • law courts • churches and other buildings of worship, crematoria • libraries open to the public, non-residential day centres, clinics, health centres and surgeries • passenger stations and termini for air, rail, road or sea travel • public toilets • zoos and menageries.
Industrial	6	Factories and other premises used for any of the following: • manufacturing, altering, repairing, cleaning, washing, breaking up, adapting or processing any article • generating power • slaughtering livestock.

As already discussed, the fire-resistant structures of the building will comply with Table B2 [below] for buildings of, ground, and upper floor levels up to 5m without a sprinkler system installed, [please see below]

Table B2 Minimum periods of fire resistance							
Purpose group of building	Minimum periods of fire resistance ⁽¹⁾ (minutes) in a:						
	Basement storey* including floor over		Ground or upper storey				
	Depth (m) of the lowest basement		Height (m) of top floor above ground, in a building or separated part of a building				
	More than 10	Up to 10	Up to 5	Up to 11	Up to 18	Up to 30	More than 30
1. Residential:							
a. Block of flats							
– without sprinkler system	90 min	60 min	30 min ¹	60 min ⁺⁵	Not permitted ⁽²⁾	Not permitted ⁽²⁾	Not permitted ⁽²⁾
– with sprinkler system ⁽³⁾	90 min	60 min	30 min ¹	60 min ⁺⁵	60 min ⁺⁵	90 min ⁺	120 min ⁺
b. and c. Dwellinghouse	Not applicable ⁽⁴⁾	30 min ¹	30 min ¹	60 min ⁽⁵⁾	60 min ⁽⁵⁾	Not applicable ⁽⁴⁾	Not applicable ⁽⁴⁾
2. Residential							
a. Institutional	90 min	60 min	30 min ¹	60 min	60 min	90 min	120 min ²
b. Other residential	90 min	60 min	30 min ¹	60 min	60 min	90 min	120 min ²
3. Office:							
– without sprinkler system	90 min	60 min	30 min ¹	60 min	60 min	90 min	Not permitted ⁽⁶⁾
– with sprinkler system ⁽³⁾	60 min	60 min	30 min ¹	30 min ¹	30 min ¹	60 min	120 min ²
4. Shop and commercial:							
– without sprinkler system	90 min	60 min	60 min	60 min	60 min	90 min	Not permitted ⁽⁶⁾
– with sprinkler system ⁽³⁾	60 min	60 min	30 min ¹	60 min	60 min	60 min	120 min ²
5. Assembly and recreation:							
– without sprinkler system	90 min	60 min	60 min	60 min	60 min	90 min	Not permitted ⁽⁶⁾
– with sprinkler system ⁽³⁾	60 min	60 min	30 min ¹	60 min	60 min	60 min	120 min ²
6. Industrial:							
– without sprinkler system	120 min	90 min	60 min	90 min	90 min	120 min	Not permitted ⁽⁶⁾
– with sprinkler system ⁽³⁾	90 min	60 min	30 min ¹	60 min	60 min	90 min	120 min ²
7. Storage and other non-residential:							
a. any building or part not described elsewhere:							
– without sprinkler system	120 min	90 min	60 min	90 min	90 min	120 min	Not permitted ⁽⁶⁾
– with sprinkler system ⁽³⁾	90 min	60 min	30 min ¹	60 min	60 min	90 min	120 min ²

Elements of structure of the building are based on the height of the uppermost habitable floor level (measured from the respective lowest floor in accordance with Diagram below: -



All elements of structure and all those supporting these elements will require a fire resistance under standard fire conditions in accordance with the relevant parts of the relevant European test methods (exposure to a fire from either side / face, or in the case of floors, underside only).

The fire strategy for the proposed project can be clarified as follows: -

- Structural fire protection will be provided in accordance with Table B2 of Approved Document B [60 minutes].
- Each floor will form a fire compartmentation floor [60-minutes],
- Each wall separating compartments will be designed as fire compartmentation walls affording at least equivalent fire protection to that of the element of structure, therefore requiring protection of 60 minutes,
- Service risers will meet the fire protection standards of the elements of structure 60 minutes, from both sides of the wall,
- Both internal and external fire spread will meet the requirements of ADB,
- Means of escape from the proposed floor levels will be initiated as follows: -
The actuation of a single smoke or heat detector head or actuation of a manual fire alarm call point with total simultaneous evacuation from all areas of the building to a place of safety,
- Travel distances on means of escape routes will comply with ADB for purpose group 5, these being 18m single direction travel and 45m where there is more than one acceptable means of escape route.
- The principles of the angle of divergence will be applied throughout the building on the basis there will be only one seat of fire and occupants can turn their backs on any seat of fire by more than 45° and move to an alternative escape route,
- Occupancy levels will be calculated on floor capacity and exit widths,
- Due to the occupancy of the prayer halls the current floor space factors contained in ADB Vol 2 do not meet the requirements to accurately calculate the occupancy levels for each area under this heading. To calculate these levels, a measurement of the prayer mats to be used has been taken which gives a calculation of 0.77m². This is a more accurate calculation than the one given in Approved Document B where the nearest calculation under this type of occupation would be 0.5m, all other areas have been calculated using Table D1.
- A fire alarm and detection system will be installed compliant with BS:5839 L2/M standards with voice alarm,
- The building will meet all requirements of Approved Document B
 - B1 - Means of Warning and Escape
 - B2 - Surface Spread of Fire - Internal Fire Spread [Linings]
 - B3 - Internal Fire Spread and Control (structure)
 - B4 - External Fire Spread

- B5 - Fire Service Requirements

1.1 Proposed Building Description

1.1.1 Ground Floor Accommodation

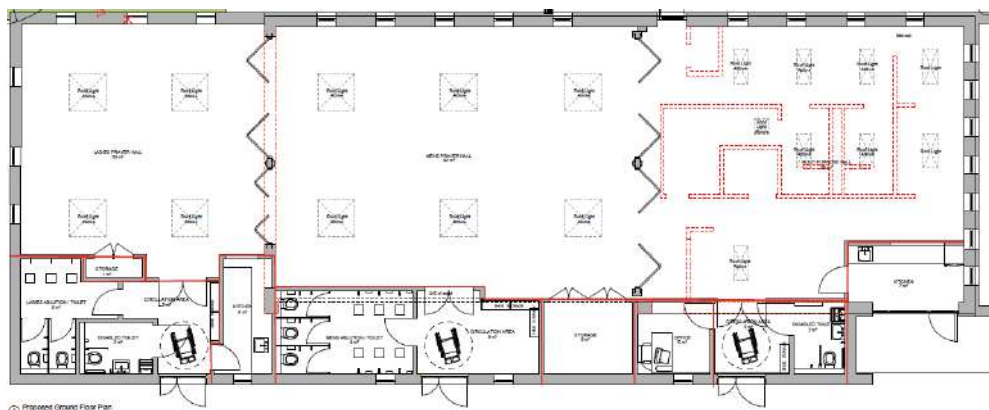
The proposed building will consist of ground floor accommodation only. The footprint of the accommodation is 148.8m², and in principle, the main area is of an 'open plan' design with protected entrance lobbies to each of the three functional areas.

The main open plan area will be sub-compartmented by bi-folding room divider doors to create: -

- Ladies Prayer Hall,
- Men's Prayer Hall,
- Multi-function Hall

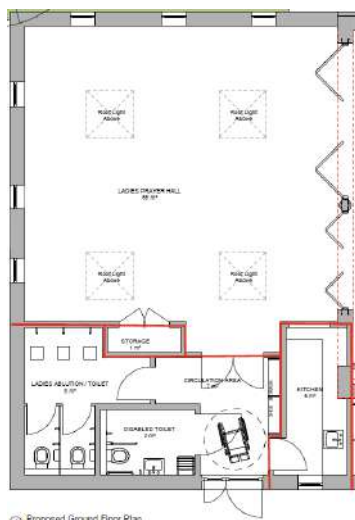
Adequate means of escape routes are provided to the front and rear elevations of the building, with a protected route to the rear of the building.

Diagram 1



30-Minute passive fire resistant structures and FD30S Fire Door Sets

Diagram 2



The Ladies Prayer Hall Area comprises of:-

Protected main entrance lobby/circulation area leading to the Ladies Prayer Hall - 6.5m²

Disabled Toilet - 3.5m²

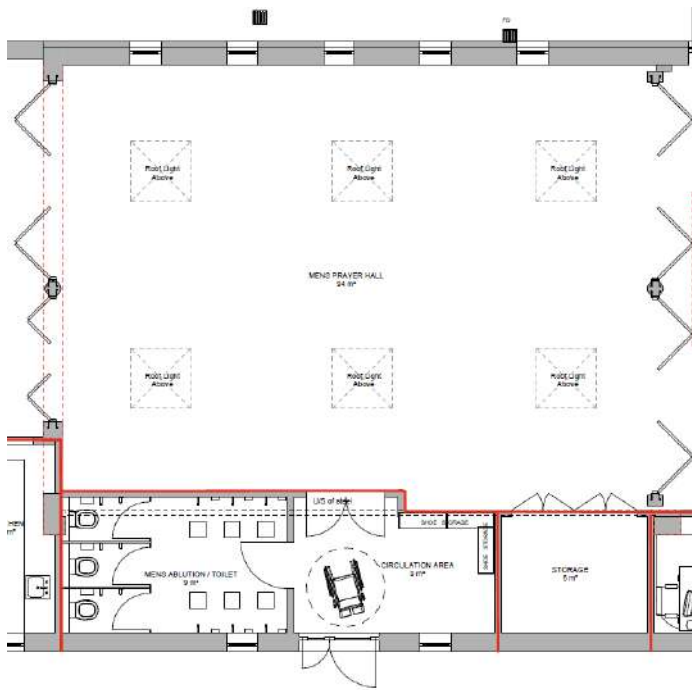
Ladies Ablution/Toilet - 7.5m²

Kitchen - 5.5m²

Ladies Prayer Hall - 57.5m²

Storage Cupboard – 1.0m²

Diagram 3



The Men's Prayer Hall Area comprises of:-

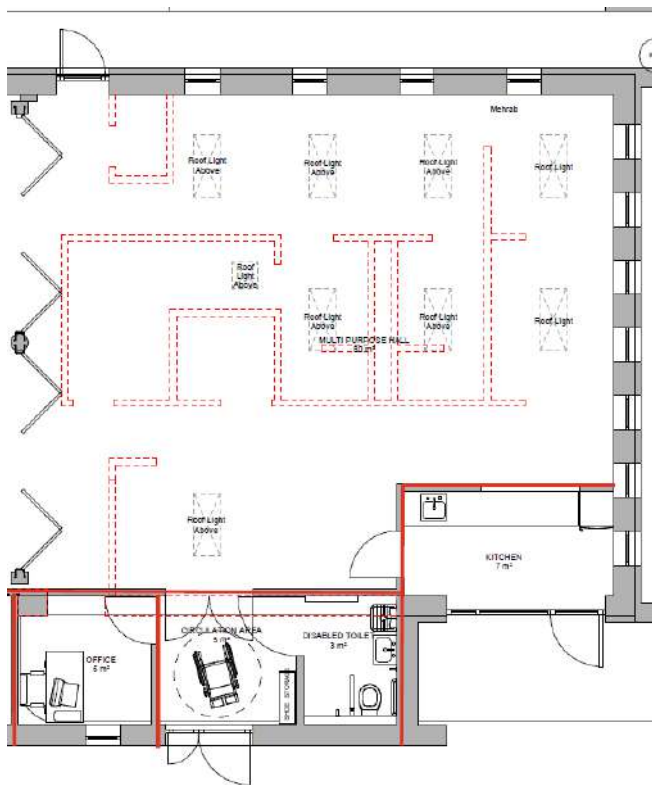
Protected main entrance lobby
lobby/circulation area leading to the
Men's Prayer Hall - 8.5m²

Men's Ablution/Toilet – 9.0m²

Men's Prayer Hall - 94.5m²

Storage Cupboard – 6.0m²

Diagram 4



The Multi-Function Hall Area comprises of:-

Protected main entrance lobby/circulation
area leading to the Multi-Function Hall –
5.0m²

Office - 5.0m²

Disabled Toilet - 3.5m²

Kitchen - 6.5m²

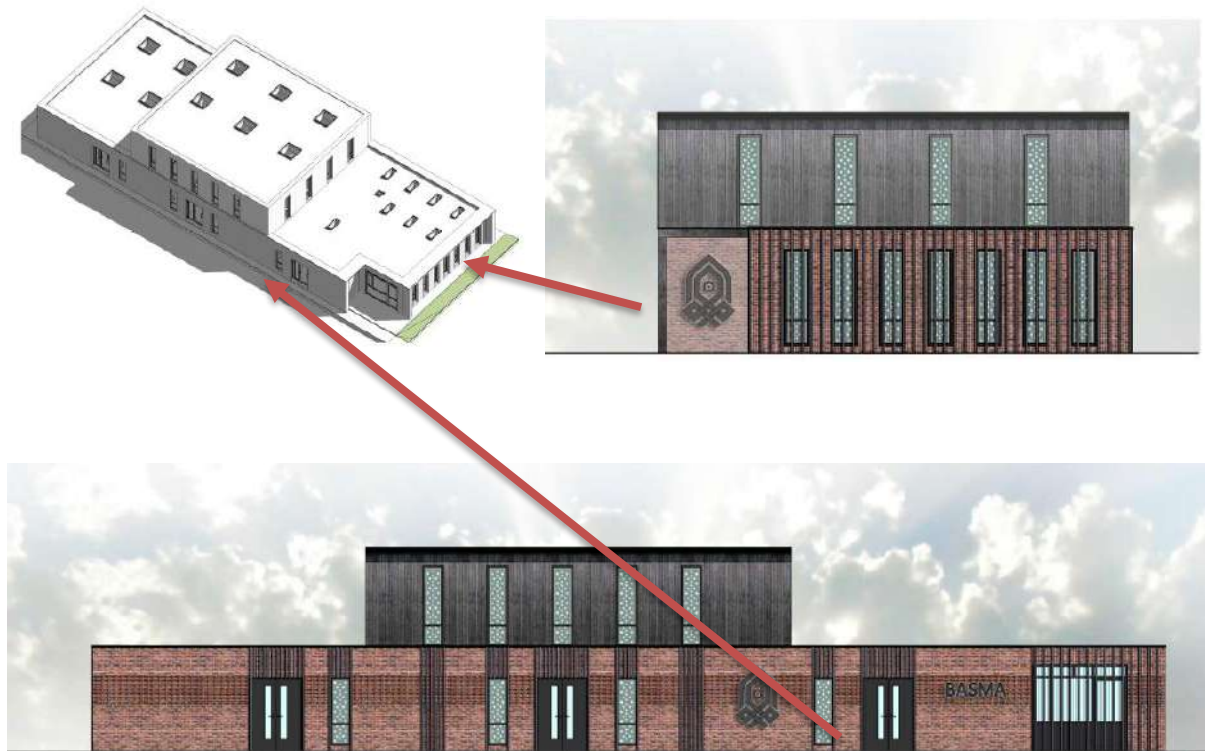
Multi-Function Hall - 80.0m²

1.1.2 Roof

The roof is constructed as a flat roof design.

Elevation Drawings

Diagram 5



1.2 Purpose Statement

This fire strategy report follows the 7 key elements of an effective Fire Strategy for Belmont and Sutton Muslim Assembly (BASMA), Former Ambulance Station, Dorset Road, Belmont, Sutton SM2 6JH these being: -

- Fire Engineering Legislation,
- Effective means of escape from all areas of the building,
- Fire Spread and its control by adequate and sufficient primary and sub-primary vertical and horizontal compartmentation,
- Construction of the building[s], including construction materials used,
- Fire Service Access to at least three sides of the building[s] and provision of fixed installations to assist the Fire & Rescue Service,
- Current fire safety management,
- Proactive fire safety management to monitor and ensure future developments can be assessed and implemented, where necessary.

The purpose of this document is to set down the fire safety objectives and the means by which these objectives for life safety and property protection are delivered at Belmont and Sutton Muslim Assembly (BASMA), Former Ambulance Station, Dorset Road, Belmont, Sutton SM2 6JH.

The proposals outlined in this document demonstrate a level of fire safety equal to or greater than standards implied by compliance with recommendations contained within Approved Document B. This level therefore satisfies the functional requirements of building regulations relating to fire safety.

This document will provide a high-level overview of the aspects of the fire safety design features that are incorporated into the building in the construction phases and subsequent proposed, occupation phase, including the management of fire safety procedures implemented on the day-to-day operation of the facility.

It is recognised that fire safety plays an important role in influencing building design. The development of fire safety objectives at an early stage of the design process and progressing throughout the project phases will ensure the continuity of cohesion of fire safety design.

For clarity, fire compartmentation and fire strategy plans are incorporated within this document. Reference will be made to the supportive floor plans of the building, identified as follows: -

- ❖ 786/BASMA/PP-200 — Proposed Site Layout Plan, Site Location Plan & Block Plan
- ❖ 786/BASMA/PP-201 — Proposed Ground Floor Plan, Birds-Eye Roof Plan and 3D View
- ❖ 786/BASMA/PP-202 — Proposed Material Elevation A
- ❖ 786/BASMA/PP-203 — Proposed Material Elevation B
- ❖ 786/BASMA/PP-204 — Proposed Sections

The fire safety strategy and design of the building reflect a single occupancy scenario.

1.3 Establishing Fire Risk Management Systems.

The risk profile for the building can be based on the type of occupancy predicted and the volume of fire loadings likely to be present within the building.

Life Risk

On this basis it can be calculated that the overall majority of life risk will be within the areas of the community centre with prayer facilities where occupants will normally be awake in unfamiliar surroundings, this life risk will be applied to all floor levels of the building within the community centre with prayer facilities demise.

The high-risk areas are adequately fire compartmented from the community centre and therefore the risk profile for this type of occupancy can be considered as people who are awake but are unfamiliar with the layout of the premises.

Fire Risk

The potential fire loadings in the community centre with prayer facilities can be calculated by existing community models and therefore can be predictable and reduced by effective fire safety management implemented by the responsible person. Equally, the same process can be considered when applying to the commercial units.

Mitigating Factors

To reduce the risk to both life and the likelihood of a fire occurring, the following are considered mitigating factors regarding provision of both passive and active fire safety measures: -

- Protected structural elements of the building,
- Passive fire compartment elements to sub-divide the ground floor level, including FD30S door sets,
- An automatic fire alarm and detection system throughout the building complying with BS5839 Part 1 Category L2/M Standards,
- Provision of Emergency Escape Lighting installed to BS5266 Standards,
- Provision of protected escape routes, include the route to the rear of the building,
- Compliant travel distances applied.

1.3.1 Fire Risk Assessment

Those responsible for the design and construction of the building will provide fire safety information to the responsible person at the completion of the development or when the building is first occupied.

1.3.2 Resources and Authority

The resources necessary to implement, maintain and improve the fire risk management system should be determined. In determining the necessary resources, account should be taken of the organisational hierarchy, the role of the fire safety manager and communication and collaboration with other users of the building. For the management of fire risk to be effective, those with fire safety responsibilities should be empowered and able to command sufficient resources to maintain the system.

1.3.3 Fire Training

The Responsible person must ensure the work force is adequately trained in all fire safety matters, including how to raise the alarm, the evacuation strategy for the building, or area, how, when, and limitations of fire extinguishers and on what type of fire they can be used on.

1.3.4 Work control

A work control system is developed reactively to work required on site to include clear lines of responsibility; a permit system; and logging and audit processes. This is to be implemented by the responsible person as designated under The Regulatory Reform [Fire Safety} Order 2005.

1.3.5 Maintenance and testing of fire safety systems

A maintenance system is one where there is monitoring of the fire safety systems, and the equipment is kept fully functional at all the times the building is in use. When systems, equipment and other arrangements are not available or not functioning correctly, alternative procedures are determined reactively. The responsible person must ensure these actions are undertaken.

1.3.6 Communications procedures

A communications system will provide information to all those involved, with alternative formats as necessary, and will have contingency arrangements.

Liaison with fire rescue services

In a system, the liaison includes arrangements for notifying the fire and rescue service of changes to the occupancy, periods of abnormal occupancy, fire growth characteristics, and other relevant factors.

1.3.7 Emergency Contingency planning

The contingency plans will include preparation and response to a wide range of foreseeable events such as the provision of shelter, communications, transport, the weather, time of day, time of week, time of year (holidays, etc.) and traffic-related issues, as well as scenarios such as power failures.

2.0 Fire safety manual

A fire safety manual will be provided and will conform to the recommendations given in Annex H of the building regulations.

2.1 Project completion

Following the final handover, the developer/management of the premises is required under current legislation to carry out a fire risk assessment.

This document will be developed following completion of the fit out works and will form part of the fire manuals developed for the premises. The fire risk assessment must be carried out by a competent person as defined within section 156 of the Building Safety Act 2022. The assessment must be made available to the Fire Authority for inspection, if required.

3.0 Regulatory Process and Standards

The building design utilised Part B of the Building Regulations as the standard to determine floor space and emergency exit strategy for the building.

In the interests of fire safety and to ensure the safety of all building users, development proposals must achieve the highest standards of fire safety and ensure that they:

- are designed to incorporate appropriate features which reduce the risk to life in the event of a fire.
- are constructed in an appropriate way to minimise the risk of fire spread.
- provide suitable and convenient means of escape for all building users.
- adopt a robust strategy for evacuation which all building users can have confidence in
- provide suitable access and equipment for firefighting which is appropriate for the size and use of the development.
- All major development proposals should be submitted with a Fire Statement, which is an independent fire strategy, produced by a third party suitably

qualified assessor. The statement should detail how the development proposal will function in terms of:

- ✚ the building's construction: methods, products and materials used,
- ✚ the means of escape for all building users: stair cores, escape for building users who are disabled or require level access, and the associated management plan approach.
- ✚ access for fire service personnel and equipment: how this will be achieved in an evacuation situation, water supplies, provision and positioning of equipment, firefighting lifts, stairs and lobbies, any fire suppression and smoke ventilation systems proposed, and the ongoing maintenance and monitoring of these
- ✚ how provision will be made within the site to enable fire appliances to gain access to the building.

It also recognises and complies with, all areas contained within **D12 of the Fire Safety London Plan** it also acknowledges the latest standards and procedures to be adopted as detailed by the Mayor of London, which states: -

“the Mayor has formally approved a new London Plan which provides further clarification about these matters and states: “to ensure that development proposals achieve the highest standards of fire safety, reducing risk to life, minimising the risk of fire spread, and providing suitable and convenient means of escape which all building users can have confidence in, applicants should consider issues of fire safety before building control application stage, taking into account the diversity of and likely behaviour of the population as a whole... Developments, their floor layouts and cores need to be planned around issues of fire safety and a robust strategy for evacuation from the outset, embedding and integrating a suitable strategy and relevant design features at the earliest possible stage, rather than features or products being applied to pre-determined developments which could result in less successful schemes which fail to achieve the highest standards of fire safety.”

D12[A] Fire Safety London Plan sets down a number of requirements as follows: -

In the interests of fire safety and to ensure the safety of all building users, development proposals must achieve the highest standards of fire safety and ensure that they:

- are designed to incorporate appropriate features which reduce the risk to life in the event of a fire.
- are constructed in an appropriate way to minimise the risk of fire spread.
- provide suitable and convenient means of escape for all building users.
- adopt a robust strategy for evacuation which all building users can have confidence in [Simultaneous Evacuation Strategy]
- provide suitable access and equipment for firefighting which is appropriate for the size and use of the development.
- All major development proposals should be submitted with a Fire Statement, which is an independent fire strategy, produced by a third party suitably qualified assessor. The statement should detail how the development proposal will function in terms of:
 - the building's construction: methods, products and materials used.

- the means of escape for all building users: stair cores, escape for building users who are disabled or require level access, and the associated management plan approach.
- access for fire service personnel and equipment: how this will be achieved in an evacuation situation, water supplies, provision and positioning of equipment, firefighting lifts, stairs and lobbies, any fire suppression and smoke ventilation systems proposed, and the ongoing maintenance and monitoring of these
- how provision will be made within the site to enable fire appliances to gain access to the building.

4.0 Fire Safety Design

The principal design is to protect the life risk within the building, which is organised around the fire safety design looking at fire growth and its resulting products of combustion, flame/heat, and smoke/gases.

The effectiveness of the building fire safety systems determines the speed, quality, and paths of movement of those products of combustion.

The ease of generation and movement of these products is influenced by the counter measures provided by the building structure.

4.1 Fire Growth

The simplest description of the fire growth process is to divide it into three fire regimes:

- (i) pre-combustion
- (ii) smouldering combustion
- (iii) flaming combustion

4.1.1 Pre-combustion is considered the process of heating fuels to their ignition point, during which time vapours and particulates are released from the fuel.

4.1.2 Smouldering is defined as glowing combustion on the fuel surface and may or may not be related to the oxygen content in the vicinity of the smouldering process.

4.1.3 Flaming combustion is almost self-explanatory, in that the production of sufficient energy and fuel vapours in the combustible range is the condition that underlines and supports the presence of flame.

4.1.4 Fire Safety Objectives

The fire safety objectives are to determine the degree to which the property and its associated facilities protect the occupants, the structure, contents, continuity of operations.

It is impossible to completely prevent the ignition of a fire in the building; therefore, the overall fire safety objective of buildings designed to a standard is a need to maintain them to that benchmark, and any subsequent alterations, modifications or refurbishments is approved in a similar manner.

Structural fire protection measures to this property have been designed in accordance with the relevant standards and Home Office guidelines. This includes both passive and active fire protection measures.

The building has been provided with the relevant level of fire rating to the structure during its construction and fit out phases. More detailed information can be obtained by referring to building plans.

5.0 Purpose Groups

The different uses within the development are classified in the following Purpose Groups under Approved Document B, this project is identified as: -

Table 1 Purpose Group[s]

Purpose group	Accommodation
<i>Purpose Group 5 – Main Centre</i>	Place of assembly, entertainment or recreation, including any of the following: • bingo halls, broadcasting, recording and film studios open to the public, casinos, dance halls • entertainment, conference, exhibition and leisure centres • funfairs and amusement arcades • museums and art galleries, non-residential clubs, theatres, cinemas, concert halls • educational establishments, dancing schools, gymnasia, swimming pool buildings, riding schools, skating rinks, sports pavilions, sports stadia • law courts • churches and other buildings of worship, crematoria • libraries open to the public, non-residential day centres, clinics, health centres and surgeries • passenger stations and termini for air, rail, road or sea travel • public toilets • zoos and menageries.

6.0 Description of Building[s]

6.1 Fire Strategy General Principles

The general philosophy for the means of escape of occupants is that there is satisfactory means of giving warning of a fire and travel distances (normally in a single direction) are limited.

Schedule 1 of the Building Regulations requires the following functional requirements to be met in respect of B1, Means of Warning and Escape:

"The building shall be designed and constructed so that there are appropriate provisions for the early warning of fire, and appropriate means of escape in case of fire from the building to a place of safety outside the building capable of being safely and effectively used at all material times.

The following discusses the implications of the proposed building designs and seeks to demonstrate that a satisfactory standard of fire safety is achieved.

The principles of an escape strategy are that in the event of a fire, people can turn their back on exposure to a fire and make their way to a place of intermediate safety (e.g., a protected route or stair core) without additional assistance from other occupants or fire fighters.

A place of intermediate safety must always give direct access to a place of ultimate safety (e.g., ground (access) level at the front, side, or rear of the building) and escaping occupants should not have to leave the place of intermediate safety [protected lobby] to reach the place of ultimate safety [outside the building]. Occupants should then be freely able to move away from the building to a place of final safety.

All Areas

- An evacuation from all areas of the community centre with prayer facilities will be initiated by either an automatic heat/smoke detector head, or by human intervention of actuating a break glass call point.
- The evacuation strategy for all areas will be simultaneous evacuation of the all occupants to an assembly point which will be located a safe distance from the building in fresh air
- Travel distances for both single direction and alternative direction [where there is more than one means of escape route] will be applied as detailed in Table 2.1.

6.2 General Recommendations

Any dead-end sections of corridor, longer than 4.5m, will be constructed as protected corridors with a minimum fire resistance of 30 minutes and FD30S self-closing doors for ADB purposed.

Where the room occupancy or floor occupancy is more than 60 persons, all doors from the individual rooms or floor will open in the direction of escape, in the case of cross-passage doors these will be FD30S doors.

In order to prevent a fire impeding two escape routes simultaneously cross-corridor separation will be introduced approximately midway between the two corridor escape routes. This separation will be a minimum of 30 minutes fire resistance with FD30S doors. If desired, it is possible for these cross-corridor doors to be held open on electro-magnetic door holders interlinked to the fire alarm in everyday use and be programmed to close on the activation of the fire alarm.

Any electronic doors, security doors or gates forming part of an escape route will need to be interfaced with the fire alarm system to disengage the door on activation of the fire alarm system or loss of power.

The escape doors secured by electronic locks [main entrance doors to the property, at ground floor level] will,

- Fail safe in open position on power failure
- Be provided with a manual release alongside the door in the form of break-glass unit (Green colour) [Type A switch]
- Be provided with a sign indicating the operation of the manual door release in an emergency.

7.0 B1 - Means of Warning and Escape

7.1 Fire Alarm and Detection System

An automatic fire alarm and detection system is to be installed to the standards of BS:5839 Part 1 Category L2/M with the addition of a voice alarm throughout the building. Further details will be submitted with design, cause and effect, and calculations regarding the fire alarm and detection system on completion of design.

Fire alarm sounders will be provided to ensure that sound pressure levels of 65dB(A) or 5dB(A) above background noise (whichever is greater) is achieved within the flat or where required, within the communal area.

All cross-corridor doors (excluding staircase doors) can be held open using magnetic hold-open devices but these should be interlinked with the fire detection system to disengage upon the activation of smoke detection. Detection coverage should then be amended from BS 5839 Part 1 to BS 7273 Part 4 and include smoke detection on either side of the associated door(s) within 0.5m – 1.5m.

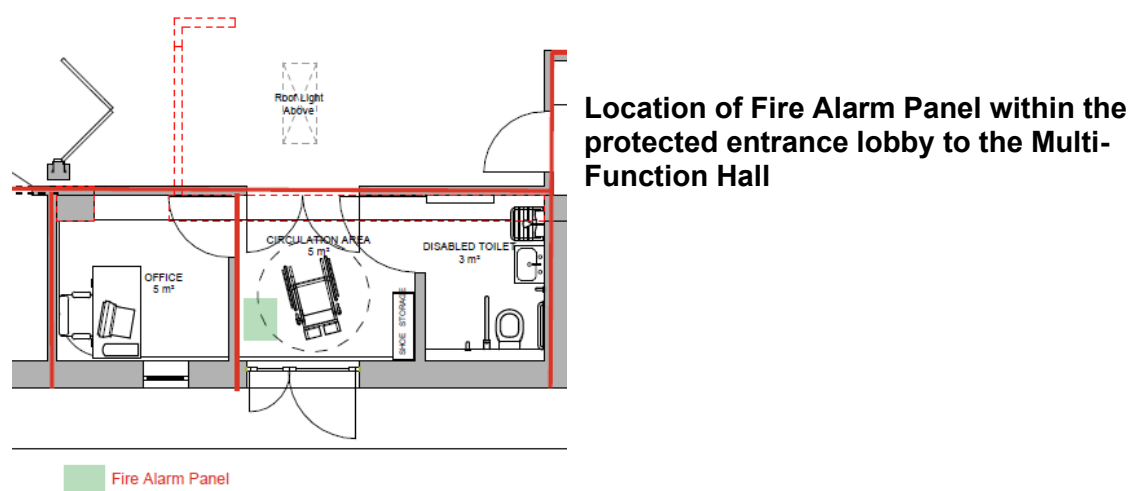
The fire alarm panel will be provided at Fire Service access level [main entrance/reception area] leading to the multi-function hall, and be visible / fully accessible by the attending fire crew upon building entry.

Further details of the automatic fire alarm and detection systems will be provided by the designated fire alarm specialist when appointed.

7.2 Table 2 - Cause and Effect

Accommodation	Fire Alarm System	Cause	Effect
Purpose Group 5 – Assembly and recreation and creche	BS5839 L2/M	Single smoke detector or manual call point	Fire Shutters operates Occupants leave the building immediately and make their way to a place of safety, externally, away from the building.

Diagram 6



7.3 Means of Escape

7.4 General Principles

The design and construction of the building ensures that, at all times, staff and visitors can move away from any fire, ultimately to a place of safety outside the building. [Total Evacuation]

1.1 The Regulatory Reform (Fire Safety) Order 2005 requires that all those features included within the approved design remain effective.

This is also covered in The Regulatory Reform [Fire Safety] Order 2005 Meaning of “general fire precautions”.

- 4.— (1) In this Order “general fire precautions” in relation to premises means,
- measures in relation to the means of escape from the premises.
 - measures for securing that, at all material times, the means of escape can be safely and effectively used.

To ensure the correct calculations are applied the following has been considered: -

- Appropriate occupancy levels
- Calculations regarding widths openings on escape routes.

7.5 Emergency Exits.

The design of all designated emergency exits must recognise the need to maintain the delicate balance between the requirements of means of escape and those of security.

Manually operated fire exit doors are provided with either push bars, or Redlam Bolts.

Any fire exit doors requiring the use of a key to release them in an emergency will be subject to re-evaluation regularly and, where possible, will be adapted to take account of technical progress.

7.6 Table 3 - Widths of escape routes and exits

Maximum Number of Persons	Minimum Width (mm)
60	750
110	850
220	1050
More than 220	5mm per person

Minimum width for an escape route should ideally be 1050mm but in any case, not less than 750mm. Please refer to fire strategy plan contained within this report.

The proposed escape routes from the building are detailed below:

Internally all fire escape routes are strategically located ensuring the angle of divergence [45°] can be maintained and the principles of their only being one seat of fire. This is demonstrated at each floor level with both single and alternative travel distances applied.

Single direction travel distances have been applied [where required] from the furthest point in any given room to either: -

- Discharging directly to open air,
- A different fire compartment.

7.7 Horizontal Exit and Escape Requirements

The requirements of horizontal exit widths depend on the type of the accommodation, size of accommodation and subsequent type of evacuation philosophy adopted.

The general recommendations are described in detail below and should adhere to Diagram C1 of Approved Document B i.e., clear width between door stop and projecting ironmongery (for doors >90° opening) or clear width between door stop and door leaf (for doors >90° opening).

The building has been designed as such that it can cater for a fully simultaneous evacuation. The building has therefore been designed with the exit widths and minimum number of exits shown in Table 6 below. Please see figure below for details on how Approved Document B requires door widths to be measured.

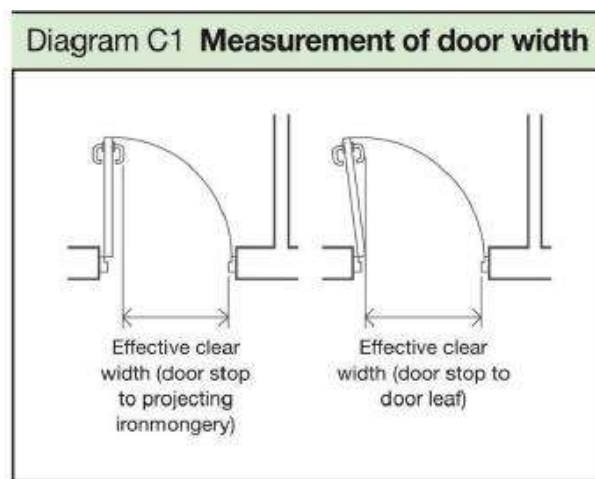


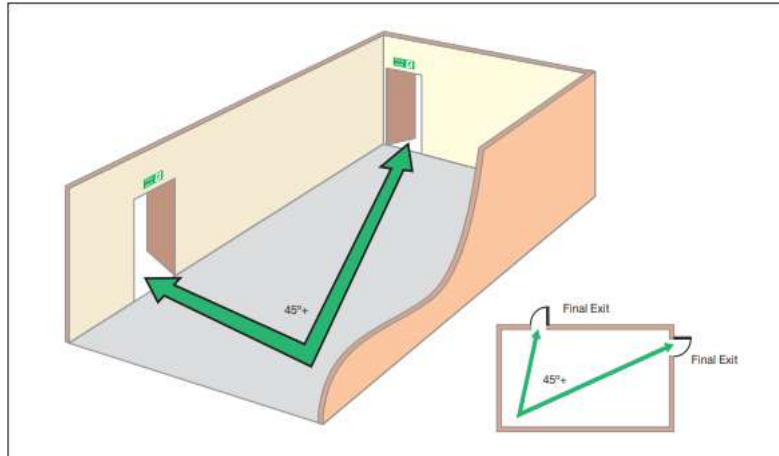
Table 4 Exit widths and minimum number of exits

Maximum number of persons	Minimum number of escape routes/exits	Minimum exit routes
0-60	1	750mm
60 -110	2	2 x 850mm
110 - 220	2	2 x 1050mm
220+	2	2 x [5mm per person]
600+	3	3 x [5mm per person]

*"Where more than 2 exits are available, it is reasonable to assume that upon a fire originating to impede a single escape route, the accommodation occupancy splits to evacuate evenly via the remaining two (or more) doors
(Subject to travel distances)"*

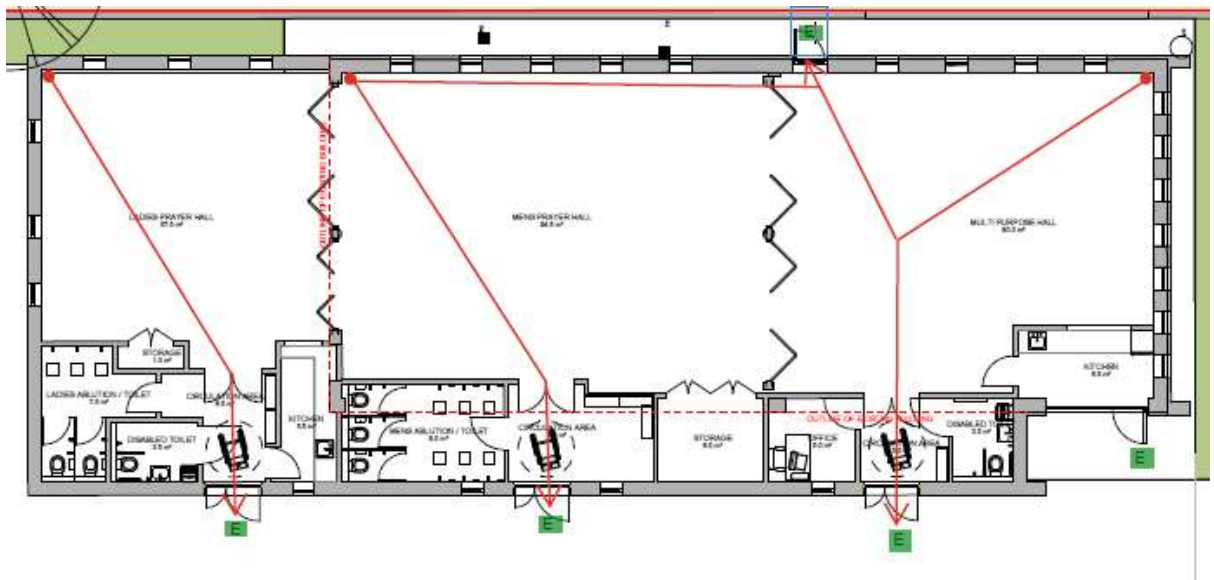
Where alternative means of escape routes have been applied, they have been based on the following conditions: -

Diagram 7



7.7.1 Ground Floor Area

Diagram 8



Adequate means of escape is provided throughout the building with all designated Fire Exits leading directly to open air and safety.

Applying single direction travel distance of 18m within the Ladies Prayer Hall is calculated to be 18m, although areas of the building alternative travel distances of 45m can be applied.

The provision of Fire Exits from the building are as follows: -

- ❖ Fire Exit leading from the Ladies Prayer Hall entrance with a width of = 1500mm
- ❖ Fire Exit leading from the Men's Prayer Hall entrance with a width of = 1500mm
- ❖ Fire Exit leading from the Multi-Function Hall entrance with a width of = 1500mm
- ❖ Fire Exit leading from the Kitchen within the Multi-Function Hall [Kitchen use only] with a width of = 750mm
- ❖ Fire Exit leading from the Multi-Function Hall rear elevation with a width of = 800mm

Using the principles of there only being one seat of fire adjacent to the largest width fire exit [worse case scenario] the building has remaining escape capabilities of 2 x 1500mm width exits and 1 x 800mm width exit.

The capabilities of the remaining exit capacity can be calculated using Table 4 as 1500mm width doors = 310 persons and 800mm width doors 60 persons

7.8 Floor Space Factors

Type of accommodation ⁽²⁾⁽³⁾	Floor space factor (m ² /person)
1. Standing spectator areas, bar areas (within 2m of serving point), similar refreshment areas	0.3
2. Amusement arcade, assembly hall (including a general purpose place of assembly), bingo hall, club, crush hall, dance floor or hall, venue for pop concerts and similar events and bar areas without fixed seating	0.5
3. Concourse or queuing area ⁽⁴⁾	0.7
4. Committee room, common room, conference room, dining room, licensed betting office (public area), lounge or bar (other than in (1) above), meeting room, reading room, restaurant, staff room or waiting room ⁽⁵⁾	1.0
5. Exhibition hall or studio (film, radio, television, recording)	1.5
6. Skating rink	2.0
7. Shop sales area ⁽⁶⁾	2.0
8. Art gallery, dormitory, factory production area, museum or workshop	5.0
9. Office	6.0
10. Shop sales area ⁽⁷⁾	7.0
11. Kitchen or library	7.0
12. Bedroom or study-bedroom	8.0
13. Bed-sitting room, billiards or snooker room or hall	10.0
14. Storage and warehousing	30.0
15. Car park	Two persons per parking space

Due to the occupancy of the prayer halls the current floor space factors contained in ADB Vol 2 do not meet the requirements to accurately calculate the occupancy levels for each area under this heading. To calculate these levels, a measurement of the prayer mats to be used has been taken which gives a calculation of 0.77m². This is a more accurate calculation than the one given in Approved Document B where the nearest calculation under this type of occupation would be 0.5m, all other areas have been calculated using Table D1. This calculation for the ground area equates to the Ladies and Men's prayer halls combined = 152 m² divided by 0.77 m² = 197 persons, with the multi-function hall measuring 80.0m² when applying the above floor space factor of 1.0 m² [please see Table D1, above] = 80 persons, with a further 20 persons occupying the office, kitchens, and welfare facilities. Total maximum capacity for the building = **297 persons**

7.9 Evacuation Procedures.

The type of evacuation proposed for this building can have a significant impact on the overall design and flexibility of the building. The evacuation strategy for this building will be simultaneous evacuation of the all occupants to an assembly point which will be located a safe distance from the building in fresh air

In this case, an evacuation from all areas of the building will be initiated either by an automatic fire detection system installed throughout the building, or human intervention by operating a break glass call point which are located on all escape routes.

7.10 Travel Distances

The travel distances from the building are on in accordance with the travel distances recommended in Approved Document B (as indicated in the table below).

Table 2.1 Limitations on travel distance				
Purpose group	Use of the premises or part of the premises		Maximum travel distance ⁽¹⁾ where travel is possible in:	
			One direction only (m)	More than one direction (m)
2(a)	Residential (institutional)		9	18
2(b)	Residential (other):			
	a. in bedrooms ⁽²⁾		9	18
	b. in bedroom corridors		9	35
	c. elsewhere		18	35
3	Office		18	45
4	Shop and commercial		18	45
5	Assembly and recreation:			
	a. buildings primarily for disabled people		9	18
	b. areas with seating in rows		15	32
	c. elsewhere		18	45
6	Industrial ⁽³⁾	Normal hazard	25	45
		Higher hazard	12	25
7	Storage and other non-residential ⁽⁴⁾	Normal hazard	25	45
		Higher hazard	12	25
2-7	Place of special fire hazard ⁽⁴⁾		9 ⁽⁵⁾	18 ⁽⁵⁾
2-7	Plant room or roof-top plant:			
	a. distance within the room		9	35
	b. escape route not in open air (overall travel distance)		18	45
	c. escape route in open air (overall travel distance)		60	100

Travel distances to be applied within the building are detailed in Table 2.1 above, according to use of the area

7.11 Emergency Escape Lighting

It is proposed to install emergency escape lighting throughout the building to conform to British Standard 5266 Part 1. Should the primary power source, the distribution panel or the local electrical wiring circuit fail, the battery back-up system will provide adequate lighting for the purposes of means of escape and health and safety.

All records and certification relating to testing and maintenance must be recorded and made available for clarification upon request.

As a generic checklist, the following areas must enjoy the benefit of Emergency Escape Lighting:

- Near Stairs or any other change of level*
- Near changes of direction and intersections of corridors*
- Illuminate Exit and other safety signs.
- Escape exits [both internal and external] – if necessary, additional lighting must be along the route towards and ultimately leading people to the assembly point to avoid panic and confusion.
- Near firefighting equipment*
- Near first aid posts*
- Open areas in excess of 60sqm
- Windowless rooms
- Lift cars
- Toilets, Lobbies and Closets exceeding 8m².
- Motor Generator, Control and Plant Rooms
- Kitchens
- Seminar Rooms – these are regarded as places of assembly generally seating more than 4 persons.

Consideration should be given for other rooms / areas where hazards or working practices dictate.

***Note:** The term near means within 2 metres measured horizontally.

Table 5.1 Provisions for escape lighting	
Use of the building or part of the building	Areas requiring escape lighting
Residential	All common escape routes ⁽¹⁾ , except in two storey blocks of flats
Office, industrial, storage and other non-residential	a. Underground or windowless accommodation b. Stairs either: • in a central core • that serve storey(s) more than 18m above ground level c. Internal corridors more than 30m long d. Open-plan areas of more than 60m ²
Shop and commercial, and car parks	a. Underground or windowless accommodation b. Stairs either: • in a central core • that serve storey(s) more than 18m above ground level c. Internal corridors more than 30m long d. Open-plan areas of more than 60m ² e. All escape routes (other than the following exception) to which the public are admitted. ⁽¹⁾ The exception is shops that meet all of the following: • have a maximum of three storeys • have no sales floor of more than 280m² • are not a restaurant or bar
Assembly and recreation	a. All escape routes ⁽¹⁾ b. Accommodation except for that which is open on one side to view sport or entertainment during normal daylight hours
Any purpose group	a. All toilet accommodation with a minimum floor area of 8m ² b. Electricity and generator rooms c. Switch room/battery room for emergency lighting system d. Emergency control rooms

Luminaries will be provided appropriately mounted (generally at least 2m in height) and spaced to provide the light levels suggested within British Standard BS 5266 Part 1 and BS EN 1838:

Table 5 Emergency Escape Lighting Requirements

Location	Illuminance	Duration
Open areas (>60m ² in area)	0.5lux	3 hours
Defined escape routes (<2m in width)	1.0lux	3 hours
Locations of disabled refuge	5.0lux	3 hours

The following provides guidance on test periods and the relevant requirements at each stage:

Daily	The Responsible Person should check central power supplies and check that indicators [red or green charging diodes] are healthy.
Monthly	The Responsible Person should perform a function test of each emergency luminaire and exit sign by simulating a mains failure for a period sufficient to check that each lamp operates from the battery, and they are all clean, in good condition and the fluorescent lamps are bright [and not blackened or flickering].
Annually [Certification required]	Testing as monthly but for the full duration [e.g., 3-hours], when it is convenient and safe to do so, and at a time of least risk. Batteries can take up to 24-hours to recharge so the test should take place when the building is not occupied or during daylight hours.

Monthly testing of the emergency lighting must be undertaken with the findings of the test recorded in the fire safety logbook. Any faults should be repaired, and the date of completion recorded. The system must also be tested to full duration and maintained annually by specialist contractors.

All records and certification relating to testing and maintenance must be recorded and made available for clarification upon request.

The system must be installed/extended in compliance with BS:5266 and tested to confirm compliance.

8.0 B2 Surface Spread of Fire - Internal Fire Spread [Linings]

The wall and ceiling surfaces in a Building can have a major influence on how fast a fire may develop. The restrictions contained in Buildings Regulations are intended to prevent the spread of fire across internal surfaces and between buildings.

The choice of materials used for wall and ceiling construction can significantly impact the spread of fire growth and is particularly important in primary circulation spaces where linings may offer the principal medium for fire spread to adversely impact the occupant escaping.

In addition, restrictions made to the internal fire spread and control can play a crucial part in the protection to firefighters in critical areas to maintain limited materials for ignitability and flammability.

For the purposes of this document the internal surface and linings do not apply to the upper surface of floors and stairs as these should not contribute to the adverse spread of flame during occupant evacuation. However, general internal spread requirements (subsequent section) refer to control of areas for firefighting activities. Doors, door frames, window frames, and frames in which the glazing is fitted, architraves and skirting are also exempt from these limitations.

The walls and ceilings should meet the recommendations shown in the table below in accordance with Approved Document B (Table 11).

Schedule 1 of the Building Regulations requires the following functional requirements to be met in respect of B2, Internal Fire Spread (Linings):

To inhibit the spread of fire within the building the internal linings shall-

- (a) adequately resist the spread of flame over their surfaces; and
 - (b) have, if ignited, a rate of heat release which is reasonable in the circumstances.
- (2) In this paragraph 'internal linings' means the materials lining any partition, wall, ceiling, or other internal structure.*

The following sections discuss the implications of the proposed building design and seek to demonstrate that a satisfactory standard of fire safety is achieved with respect to both requirements stated above.

Table 6.1 Classification of linings	
Location	Classification
Small rooms of maximum internal floor area: a. 4m ² in residential accommodation b. 30m ² in non-residential accommodation	D-s3, d2
Other rooms (including garages)	C-s3, d2
Other circulation spaces	B-s3, d2(1)
NOTE: 1. Wallcoverings which conform to BS EN 15102, achieving at least class C-s3, d2 and bonded to a class A2-s3, d2 substrate, will also be acceptable	

It should be ensured that, wherever possible, all furniture, fixtures and fittings have a low ignitability, are of limited combustibility, and do not give off noxious fumes if ignited.

The Classification of linings recommended in the table above can be downgraded (but not less than Class 3 or D-S3, d2) in walls of rooms providing the total area of those parts in any one room does not exceed one half of the floor area of the room and subject to a maximum of 20m² in residential accommodation and 60m² in non-residential accommodation.

8.1 General Internal Fire Spread Requirements

Non-combustible and limited combustible construction material should be used in the following situations.

- Flooring and floor coverings within stair cores should be of a low flammability to limit spread of flame. Textile coverings should be tested, with underlay, to BS 4790 to ensure non- or limited ignitability classification. Floor coverings will be secured with

non-water-based adhesive and be interrupted at all door thresholds with non-combustible strips of a minimum of 50mm.

- Furniture and fittings can have a major impact on fire spread but is not covered within the Building Regulations through Approved Document B. It is however recommended to keep display materials and decorations (permanent and seasonal) to a minimum with ongoing fire risk assessment, by the Building Management, to maintain a permanent pathway of means of escape and limited impact to the defined areas of sterility when the building is in use.
- Further guidance should be sought from the Furniture and Furnishings (Fire Safety) Regulations with further input from an interior designer.
- The internal design of the building is based on providing adequate passive fire compartmentation to reduce the spread of fire internally, protected all means of escape routes, separate high hazard areas from normal risk areas, and reduce travel distances down to prescribed acceptable levels. This is achieved using the following principles: -
 - ❖ Walls built of either brick or stud-work covered with fire resisting boarding and taped and jointed and adequately fire stopped at all junctions using appropriate fire stopping materials. Where cabling or pipework transverse the structure[s], appropriate fire stopping measures must take place.
 - ❖ The walls must offer adequate fire resistance from both sides [according to location, as detailed in the document], any coverings must meet the standard detailed in Table 6.1 [above]
 - ❖ Fire doors are to be provided where required to maintain passive fire compartmentation and support this strategy. The door[s] should be self-closing and when fitted in its frame be capable of meeting the performance requirements for a FD30S/60S door i.e. a fire door assembly which would satisfy the integrity criterion for 30/60 minutes if tested in accordance with BS EN 1634-1:2000 and restrict the passage of smoke at ambient temperature to a leakage rate not exceeding $3\text{m}^3/\text{m}/\text{hour}$ [head & jambs only] when tested at 25pa under BS:EN13501." An approved self-closing device must be fitted to the door[s].
 - ❖ Any glazing within the fire compartmentation structures will meet the standards of **EW-Class. Design definition** - As well as stopping flame, smoke and gases EI Class systems act as a barrier to radiant heat and reduces the radiant energy to $1\text{KW}/\text{m}^2$ at 1m from the glazing. This means you could stand on the glass directly opposite the fire and feel no increase in heat, giving complete fire safety from the glazed element. All ratings are described in terms of insulation or integrity length of time from 30 minutes to 2 hours (120 minutes).
 - ❖ Ceilings will also meet the required standards of fire resistance and fire spread as detailed in Table 6.1.

9.0 B3 Internal Fire Spread and Control (structure)

Table B2 Minimum periods of fire resistance								
Purpose group of building	Minimum periods of fire resistance ⁽¹⁾ (minutes) in a:							
	Basement storey* including floor over		Ground or upper storey					
	Depth (m) of the lowest basement		Height (m) of top floor above ground, in a building or separated part of a building					
	More than 10	Up to 10	Up to 5	Up to 11	Up to 18	Up to 30	More than 30	
1. Residential:								
a. Block of flats								
– without sprinkler system	90 min	60 min	30 min ¹	60 min ⁺⁵	Not permitted ⁽²⁾	Not permitted ⁽²⁾	Not permitted ⁽²⁾	
– with sprinkler system ⁽³⁾	90 min	60 min	30 min ¹	60 min ⁺⁵	60 min ⁺⁵	90 min ⁺	120 min ⁺	
b. and c. Dwellinghouse	Not applicable ⁽⁴⁾	30 min ¹	30 min ¹	60 min ⁽⁵⁾	60 min ⁽⁵⁾	Not applicable ⁽⁴⁾	Not applicable ⁽⁴⁾	
2. Residential								
a. Institutional	90 min	60 min	30 min ¹	60 min	60 min	90 min	120 min ²	
b. Other residential	90 min	60 min	30 min ¹	60 min	60 min	90 min	120 min ²	
3. Office:								
– without sprinkler system	90 min	60 min	30 min ¹	60 min	60 min	90 min	Not permitted ⁽⁶⁾	
– with sprinkler system ⁽³⁾	60 min	60 min	30 min ¹	30 min ¹	30 min ¹	60 min	120 min ²	
4. Shop and commercial:								
– without sprinkler system	90 min	60 min	60 min	60 min	60 min	90 min	Not permitted ⁽⁶⁾	
– with sprinkler system ⁽³⁾	60 min	60 min	30 min ¹	60 min	60 min	60 min	120 min ²	
5. Assembly and recreation:								
– without sprinkler system	90 min	60 min	60 min	60 min	60 min	90 min	Not permitted ⁽⁶⁾	
– with sprinkler system ⁽³⁾	60 min	60 min	30 min ¹	60 min	60 min	60 min	120 min ²	
6. Industrial:								
– without sprinkler system	120 min	90 min	60 min	90 min	90 min	120 min	Not permitted ⁽⁶⁾	
– with sprinkler system ⁽³⁾	90 min	60 min	30 min ¹	60 min	60 min	90 min	120 min ²	
7. Storage and other non-residential:								
a. any building or part not described elsewhere:								
– without sprinkler system	120 min	90 min	60 min	90 min	90 min	120 min	Not permitted ⁽⁶⁾	
– with sprinkler system ⁽³⁾	90 min	60 min	30 min ¹	60 min	60 min	90 min	120 min ²	

9.1 Statutory Requirements

The building shall be designed and constructed so that, in the event of fire, its stability will be maintained for a reasonable period and so that the unseen spread of fire and smoke within concealed spaces in its structure and fabric is inhibited.

9.2 General Principles

The building will be constructed with consideration to resistance to collapse, fire penetration and the transfer of heat to ensure that elements of loadbearing structure are capable to withstand the effects of fire for an appropriate period without loss of stability.

Consideration will also be given to the restriction of fire spread throughout the construction by way of sub-dividing the building into fire separated compartments. The primary objectives of compartmentation are to both prevent rapid spread of fire to impact on occupant evacuation and reduce the likelihood of potentially severe fires (to other occupants and firefighters).

Further thought should also be provided against unseen fire spread in cavities to limit concealed danger to both occupants evacuating and firefighters tackling a fire; unaware of the further potential risk in immediate hidden areas.

9.3 Loadbearing Elements of Structure

Elements of structure are loadbearing members to the primary structure of the building i.e. structural framing systems, beams, columns etc. Where one element of structure supports or gives stability to another, the supporting element should have no less fire resistance and protection than the element it supports. The measures also provide for elements of structure that are common to more than one building or compartment to be constructed to the relevant provisions.

Any elements of structure which only support themselves or a roof do not require any specific fire resistance.

Elements of structure of the development will be based on the height of the top floor level (measured from the respective lowest floor in accordance with Diagram C6 of Approved Document B). All elements of structure and all those supporting these elements will require a fire resistance shown in table below, under standard fire conditions in accordance with the relevant parts of BS:EN13501 (exposure to a fire from either side / face, or in the case of floors, underside only).

External walls should be considered as elements of structure if designed as loadbearing to the primary structure. Further requirements for external wall fire resistance (if any) are given in this report for external fire spread control.

Schedule 1 of the Building Regulations requires the following functional requirements to be met in respect of B3, Internal Fire Spread (Structure):

- (1) The building shall be designed and constructed so that, in the event of fire, its stability will be maintained for a reasonable period.*
- (2) A wall common to two or more buildings shall be designed and constructed so that it adequately resists the spread of fire between those two buildings.*
- (3) To inhibit the spread of fire within the building, it shall be sub-divided with fire resisting construction to an extent appropriate to the size and intended use of the building.*
- (4) The building shall be designed and constructed so that the unseen spread of fire and smoke within concealed spaces in its structure and fabric is inhibited.*

The following sections discuss the implications of these requirements on the proposed design of the building.

9.4 Compartmentation and Fire Spread

The Building Regulations Approved Document B requires that the building needs to be compartmented using fire resisting construction to floors and walls. This is also covered in The Regulatory Reform [Fire Safety] Order 2005 Meaning of “general fire precautions”.

- 4.— (1) In this Order “general fire precautions” in relation to premises means,
- measures to reduce the risk of fire on the premises and the risk of the spread of fire on the premises.
 - measures in relation to the means of escape from the premises.
 - measures for securing that, at all material times, the means of escape can be safely and effectively used.

The objective of this is: -

- a. to provide sufficient resistance to the effects of fire preventing rapid fire spread which could trap occupants and maintaining structural stability to provide adequate time for escape and extinguishment,
- b. to inhibit the spread of fire and smoke within the building thereby reducing the chances of a fire becoming large, as large fires represent a greater hazard to building occupants,
- c. to inhibit the spread of fire to adjacent buildings
- d. to reduce the fire threat to fire fighters and other persons in the immediate vicinity

To ensure that life safety and assets can be protected, fire compartments have been constructed to limit the spread of fire within the building, with each fire compartment kept within the limits of those stated within building regulations and, where applicable, to separate the higher risk areas from the lower risks and contain any potential high fire loads.

9.5 Internal subdivision

The way a building is subdivided internally affects the risk to users and their ability to use the planned means of escape in the event of a fire. The property meets all requirements regarding subdivision, as detailed in Building Regulations.

9.6 Compartments

The building has been divided in different compartments with escape staircases per floor, giving to each compartment an assigned means of escape of one direction, reducing the number of occupants to evacuate from the building, increasing the security and feasibility of the project.

The building will not store dangerous substances or preparations such as substances or preparations that have a fast or ultra-fast fire growth rate or are classified as explosive, oxidizing, extremely flammable, highly flammable under the Chemicals (Hazard Information and Packaging for Supply) Regulations 2002 [26], are stored and/or used, the area is considered to be of high fire risk.

9.7 Methods of Fire Rating

The fire resistance periods for compartmentation are based on complying with the requirements of the Building Regulations, as appropriate at the time of construction.

Subsequent other newer versions of the regulations are applied where modifications or extensions have been or are made to the structure.

Compliance with the recommendations of British Standards relating to structural fire precautions and fire safety installations are also applied.

The compartment strategy consists of passive fire protection measures implemented by means of fire resisting walls, fire resisting doors, fire stopping, to meet the current standards and appropriate materials used with the correct fire ratings as those listed below: -

- a) Load bearing steel work must be protected by a minimum of 60 minutes fire resistance.
- b) fire resisting block work and stud partitions enclosing the staircase enclosures at each floor level,
- c) fire resisting block work and stud partitions enclosing the hazard risk rooms from other areas [storerooms and Plantrooms],
- d) fire resisting block work and stud partitions forming sub-compartments to reduce single direction travel distances to acceptable levels,
- e) Sub-compartments formed by walls having a minimum period of fire resistance of 60/30 minutes, either terminated at the underside of a compartment floor, or a ceiling that is non-demountable and imperforate and has a minimum period of fire resistance of 60 minutes when tested from below in accordance with BS: EN: 13501.
- f) Service Risers must have a minimum period of fire resistance of 60 minutes construction with 60 minutes fire-resisting door-sets, when tested in accordance with BS:EN 13501. The doors must display signage 'Fire Door Keep Locked' at eye level on the front face of the door.
- g) fire resisting timber doors, [FD60/30S] incorporating intumescent and smoke seals, in the partitions/walls, fitted with either self-closing devices or withhold-open devices actuated by the fire warning system as appropriate, or fitted with a lock to be kept locked shut when not in use,
- h) fire stopping provided to all compartment wall junctions with external walls, compartment floors.
- i) All ductwork which transverses passive fire compartments must have fire dampers installed within the ductwork at the appropriate locations. The fire dampers must be of the equivalent fire rating as the passive fire construction. These must be installed by a competent person and compliance certificates issued.
- j) Any breaches to allow the passage of cabling in either vertical or horizontal fire compartmented structures must be adequately fire stopped by a competent contractor and compliance certificates issued for each breach.
- k) Where breaches to allow the passage of pipework in either vertical or horizontal fire compartmented structures are needed approved fire collars are to be used and adequately fire stopped by competent contractor and compliance certificates issued for each breach.

9.8 Fire Doors

Fire doors must also be integral and remain free from obstruction at all times.

Fire doors are required to provide two main functions:

A] To maintain any compartmentation of the buildings, this has been introduced to limit the size and spread of fire in order to control the perceived risk.

B] To allow access to protected escape routes, both vertically and horizontally, without any loss of fire resistance, and limit smoke movement in the structure forming these routes, i.e., protected corridors and protected shafts [e.g., staircases and riser cupboards].

A full survey of all fire resisting doors must be undertaken on a monthly basis to ensure door and door-set meet the latest standards.

The visual survey must include:

- No signs of damage which could reduce the integrity of the door or doorframe.
- The door is hung on three approved hinges.
- Intumescent strips and smoke seals are correctly fitted.
- The door is hung correctly to ensure correct gaps are maintained between door frame and door[s]
- Approved self-closing devices are fitted and the doors close effectively to form an effective seal, or close effectively onto the doorstops contained within the doorframe.
- All fire resisting doors are correctly identified with compliant signage.

To be effective a fire door must close firmly and securely into the frame and be held there until the intumescent seal is activated, leaving no escape route for flames or deadly smoke. And it must do this every time the door is opened.

A definition and specification of the standards is stated below:

[Latest standards of fire resisting door sets] *“The door[s] should be self-closing and when fitted in its frame be capable of meeting the performance requirements for a FD30S/60S door i.e. a fire door assembly which would satisfy the integrity criterion for 30/60 minutes if tested in accordance with BS EN 1634-1:2000 and restrict the passage of smoke at ambulant temperature to a leakage rate not exceeding 3m³/ m/ hour [head & jambs only] when tested at 25pa under BS:EN13501” An approved self-closing device must be fitted to the door[s].*

The smoke seals are designed to provide additional protection and compliment the intumescent strip and to prevent the passage of cold smoke. All fire door-sets should have fitted intumescent strips and cold smoke seals [brushes are a lot more robust and durable than plastic neoprene] on all 3 edges of the door, or preferably the 3 edges of the door frame itself and along the middle where the two doors meet. The ambient smoke seals and intumescent strips should be continuous and run the full length of the door edge i.e., not short lengths or cuts to fill gaps.

See below diagrams explaining where the fitted intumescent strips and ambient smoke seals can be placed to ensure sufficient compartmentation is achieved:

Diagram 9



There should be no more than a constant of a gap than 3mm + 1mm [4mm] or 3mm – 1mm [2mm] from the frame to door leaf. Where double doors are used, the same gap distancing method should be used where the two doors meet.

The gap between the base of the door and the floor should be no more than 3mm. [Intumescent strips and smoke seals are not required on the bottom edge of the door leaf].

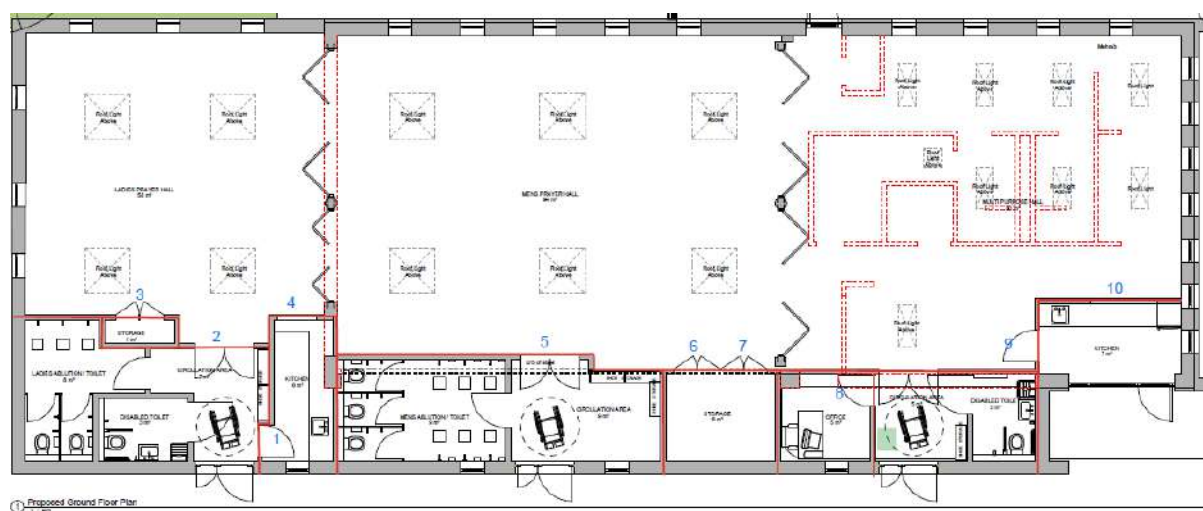
An effective Fire Door set must also have 3-hinges and self-closing mechanisms where needed [self-closing mechanisms are not necessarily required for fire doors on storerooms with an occupancy type such as the one being inspected; however, consideration should be given to the working practices of the business as to whether they are required or not].

If painting of fire doors is essential, careful consideration must be made and consultation if needed to ensure non-flammable paints are utilised which will not compromise the effectiveness of the fire door set in the event of a fire.

Regular and routine checks [monthly] of all fire doors should be completed and records kept. It is recommended that an itinerary of fire door-sets is maintained including door numbering. This will prove beneficial when completing a maintenance programme of fire door-sets. If fire doors are required to remain open, then a fire safety solution must be sought to ensure compartmentation is not compromised.

A closed fire door can stop fire in its tracks, but wedged open, the fire door provides no protection, and the fire, smoke and toxic fumes can spread throughout the building. There are a number of solutions available; including door holding devices [Briton 996] or electro-magnetic door holding devices, both of these systems to be interfaced with the Fire Alarm system.

Diagram 10



Fire Door Provision Ground Area

Table 6

Location – Floor Level	Internal Location	Fire Resisting Standard	Approved self-closing device fitted	Door Signage
Ground Floor Area Door 1	Single door leading from the protected lobby area and Ladies Hall to the kitchen	FD30S	Yes	 Installed on both faces of the door at eye level
Ground Floor Area Door 2	Double cross-passage doors separating the protected lobby from the Ladies Prayer Hall	FD30S	Yes	 Installed on both faces of the door at eye level
Ground Floor Area Door 3	Double doors to the storage cupboard within the Ladies Prayer Hall	FD30S	No	 Installed front face of the door at eye level

Ground Floor Area Fire Shutter 4	Fire Shutter located within the fire compartmentation wall between the kitchen and the Ladies prayer hall	FD30	Interfaced with the fire alarm	
Ground Floor Area Door 5	Double cross-passage doors from the Men's Prayer Hall to the protected lobby area	FD30S	Yes	 Installed on both faces of the door at eye level
Ground Floor Area Door 6	Double doors to the storage cupboard within the Men's Prayer Hall	FD30S	No	 Installed front face of the door at eye level
Ground Floor Area Door 7	Double doors to the storage cupboard within the Men's Prayer Hall	FD30S	No	 Installed front face of the door at eye level
Ground Floor Area Door 8	Single door leading from the protected lobby area leading to the multi-function hall to the office	FD30S	Yes	 Installed on both faces of the door at eye level
Ground Floor Area Door 9	Single door leading from the multi-function hall to the kitchen	FD30S	Yes	 Installed on both faces of the door at eye level
Ground Floor Area Fire Shutter 10	Fire Shutter located within the fire compartmentation wall between the kitchen and the multi-function hall	FD30	Interfaced with the fire alarm	

Ground Floor Area Door 11	Double cross-passage doors from the Men's Prayer Hall to the protected lobby area	FD30S	Yes	 Installed on both faces of the door at eye level
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9.9 Fire Compartmentation

All floors within the project will be constructed as compartment floors achieving the equivalent protection to that of the element of structure i.e. 60 minutes.

Due to the sleeping risks within the accommodation, the common corridor[s] should be designed as a protected corridor achieving 30 minutes fire rated construction with FD30S fire rated doors (self-closing) provided as entrance doors to each individual flat.

All service risers will be fire stopped horizontally with 60 minutes fire resistance between ground and upper floor levels and include 60-minute enclosures throughout the vertical shaft work. Doors will achieve a minimum of FD60S.

9.10 Cavity Barrier Protection

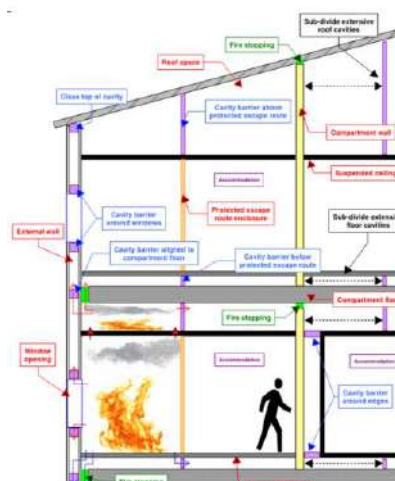
Cavity barrier protection outlines the requirements for the restriction of smoke and flames through sub-dividing cavities for the purpose of either cavities representing a potential pathway for spread of fire to impact on means of escape and firefighters or extensive cavities signifying possibly severe unseen fire spread.

The key areas that require cavity barriers are as follows:

Cavity barrier provision should be designed under Diagram 33 of Approved Document B as represented in the Diagram 11 of this report] below.

- At the junction between an external cavity wall and a compartment wall that separates buildings,
- At the top of such an external cavity wall,
- At the junction between an external cavity wall and every compartment floor / other compartment wall,
- At the junction between an internal cavity wall and every compartment floor, compartment wall, or other wall or door assembly that forms a fire resisting barrier,
- In a protected escape route, above and below any fire resisting construction that is not carried full storey height,
- Within the void behind the external face of rain screen cladding at every floor level and on the line of compartment walls abutting the external wall, and
- At the edges of cavities (including around openings i.e. windows)

Diagram 11



In addition to the above locations, cavity barriers are required in cavities (including ceiling voids and under floor service voids) where the cavity exceeds 10m - 20m in any direction.

Table 7 Cavity Barrier Requirements

Location of cavity	Class of surface / product exposed in cavity (excluding the surface of any pipe, cable or conduit, or any insulation to any pipe)		Maximum dimension in any direction
	National Class	European Class	
<i>Between roof and a ceiling</i>	<i>Any</i>	<i>Any</i>	<i>20m</i>
<i>Any other cavity</i>	<i>Class 0 or Class 1</i>	<i>Class A1 Class A2-s3, d2 Class B-s3, d2 Class C-s3, d2</i>	<i>20m</i>
<i>Any other cavity</i>	<i>Not Class 0 or Class 1</i>	<i>Not any other above classes</i>	<i>10m</i>

In the event that larger cavities are required a summary of the necessary provisions are discussed below:

Table 8 Approved Document B Recommendations on Cavity Barriers

Approved Document B recommendations from Paragraph 9.12
A) The room and the cavity together are compartmented from the rest of the building.
B) An automatic fire detection and alarm system meeting the relevant recommendations of BS 5839 Part 1 is fitted in the building (however detectors are not required in the cavity – subject to void protection under BS 5839 Part 1)
C) The cavity is used as a plenum and the recommendations about re-circulating air distribution systems in BS 9999 are followed.
D) The surface of the ceiling exposed in the cavity is Class 0 and the supports and fixings in the cavity are non-combustible construction.
E) The flame spread rating of any pipe insulation system is Class 1 .
F) Any electrical wiring in the void is laid in metal trays, or in metal conduit.
G) Any other materials in the cavity are of limited combustibility.

The cavity barriers will provide a 30-minute fire rating (i.e. 30 minutes integrity and 15 minutes insulation).

Cavity barriers in a stud wall / partition or provided around openings may also be formed using construction materials (rather than specified products) of the following:

- Steel at least 0.5mm thick,
- Timber at least 38mm thick,
- Mineral wool (compressed), or
- Calcium silicate / cement-based / gypsum-based boards at least 12mm thick.

Cavity barriers should be either be fitted under compression or mechanically fixed in position.

9.11 Fire Barriers and fire stopping

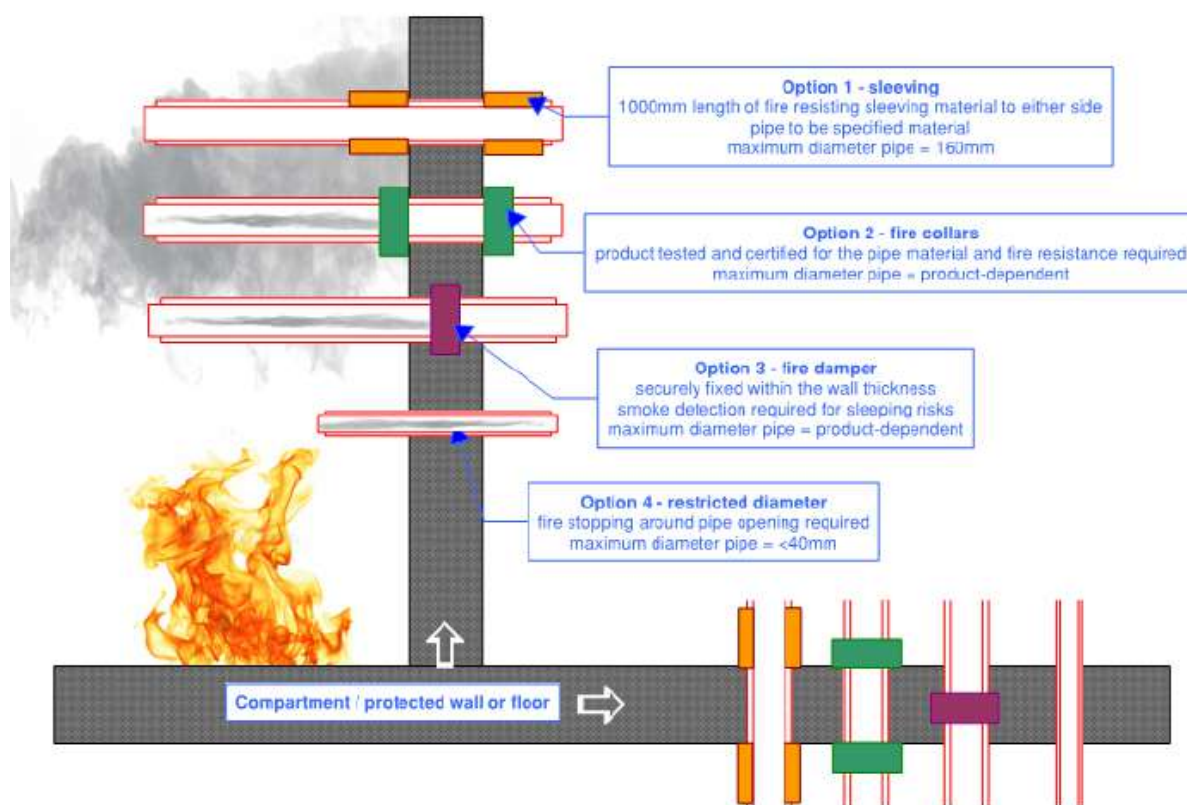
Dissimilar to cavity barrier protection, fire stopping requirements consist of maintain the fire compartmentation of the wall or floor in which it resides. Therefore, the fire resistance of the material / product specified must be achieve the fire resistance required for the associated compartment line it protects. Every joint to another element, imperfection of fit or opening to allow services must be adequately protected by sealing or fire stopping so that the resistance of the element is not impaired.

Any openings for services (exceeding a dimension set out in the Table of Approved Document B [shown below] breaching compartment walls will be fire stopped (unless protected throughout their entire length with fire resistant material) in accordance with Section 10 of Approved Document B as outlined in the following:

- **Option 1 – Sleeving:** A pipe of lead, aluminium (or alloy), uPVC, or fibre-cement with a maximum nominal diameter of 160mm with fire resistant sleeving on both sides of the associated wall of not less than 1000mm. Sleeving (or non-combustible pipe) to be in contact with the service pipe.
- **Option 2 – Fire collars (proprietary sealing):** A service pipe (of any material and nominal diameter) with a specified product, certified for the application used to maintain the resistance of the compartment line.
- **Option 3 – Fire dampers:** Securely fixed, automatic fire rated damper within the service opening to close in the event of a fire (thermally actuated devices). Smoke detection-actuation to be used where a sleeping risk is evident.
- **Option 4 – Restricted internal diameter:** Maintain the opening as small as practicable given the pipe material used as per Table 14 with appropriate fire stopping material around the service opening.

Typical representation of fire stopping to service penetrations is shown in the diagram below:

Diagram 12 Typical Representation of Fire Stopping Protection



This is primarily to prevent the passage of fire and assist in retarding the movement of smoke across the lines of compartmentation.

Table 9 Fire Stopping Requirements

Situation	Pipe material and maximum nominal internal diameter		
	Non-combustible material	Lead, aluminium (or alloy), uPVC, fibre cement	Any other material
Structure (but not a wall separating buildings) enclosing a protected shaft which is not a staircase or a lift shaft	160mm	110mm	40mm
Compartment wall or compartment floor between flats	160mm	160mm (stack pipe) ¹ 110mm (branch pipe) ¹	40mm
Any other situation	160mm	40mm	40mm

¹Relevant to above drainage systems only – in all other cases, 40mm to be applied

Products used for fire stopping and sealing systems should be certified as tested in the application intended. However general fire stopping materials can also be used:
Cement mortar,

Gypsum-based plaster,
Cement-based or gypsum-based vermiculite or perlite mixes,
Glass fibre, crushed rock, blast furnace slag or ceramic-based products, or
Intumescent mastics.

10.0 B4 External Fire Spread

Schedule 1 of the Building Regulations requires the following functional requirements to be met in respect of B4, External fire spread:

- (1) The external walls of the building shall adequately resist the spread of fire over the walls and from one building to another, having regard to the height, use and position of building.
- (2) The roof of the building shall adequately resist the spread of fire over the roof and from one building to another, having regard to the use and position of the building.

The external fire spread for all roof coverings must meet the current standards of classification for the roofs

Approved document B4 stipulates the following: -

14.1 'Roof covering' describes one or more layers of material, but not the roof structure as a whole.'

10.1 General Principles

The design for external walls, extent of unprotected facades (per associated compartment) and roof classification can play a key part in restricting the severity of fire spread either across the external fascia of the building or onto a neighbouring boundary line / building.

10.2 External Wall Construction

The external envelope of the building should not be a medium for fire spread under Approved Document B.

External walls considered as loadbearing to the primary structure of the building should achieve a fire rating equivalent to the element of structure fire resistance value for the associated block (unless forming an unprotected area defined within this report).

External walls should adhere to BR 135 for cladding systems of multi-storey buildings over 18m in height using full scale test data from BS 8414 Parts 1 or 2 or include insulation product filler material of limited combustibility. This restriction need not apply to masonry cavity wall construction which complies with Diagram 34 of Approved Document B by way of two leaves of brick or concrete, each at least 75mm thick (internal and external of the cavity).

In any circumstance where the proposed external wall system differs from a BS 8414 tested system, further assessment should be carried out to review the impact of the amended detail.

Should it be confirmed that materials of combustibility are proposed or amendments to a tested system are required, guidance is advised through the Building Control Alliance (BCA) Technical Note 18 with independent documentation to this report assessing the design intent.

From information supplied the external walls will be of block/brickwork with external finishes of : -

Option A

- 1 / Timber Geometric Screen
- 2 / Charred Timber ShiguBan Cladding
- 3 / Timber Slats
- 4 / Patterned Red Brick
- 5 / Green Climbing Plants

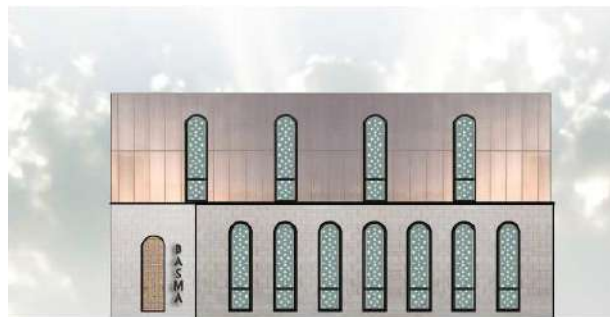


BASMA FRONT MATERIALITY ELEVATION - OPTION A

Or

Option B

- 1 / White / Cream Stone
- 2 / Bronze Cladding
- 3 / Red and Beige Stone Detailing



BASMA FRONT MATERIALITY ELEVATION - OPTION B

The external wall system [EWS] will comply with building regulations for buildings under 18m in height to a minimum standard which requires non-combustible cladding of a **C-s3, d2** or better.

To clarify the above, **C** indicates the material has a limited contribution to a fire, **s3** refers to smoke development where it produces a substantial amount of smoke and **d2** refers to droplets produced when burning, which is considered a large amount in the first 10 minutes of exposure.

10.3 Space Separation

Any extent of the cross-sectional area of a compartment along the external wall line is considered to be an unprotected area for external fire spread unless provided with fire resistance. For the purpose of this assessment, external walls of protected shafts (considered as sterile areas) are excluded.

The extent of the requirements of fire resistance to the unprotected areas (if any) are governed by the height and width of the compartment and the distance from the compartment to the relevant boundary line. This forms an assessment of the likelihood of fire spread to occur to buildings on an adjacent site or to separate fire compartments on the same development.

Calculations have been carried out for the required boundary distances and achieved boundary distances based on the largest compartment façade / worst-case scenario of each relevant elevation. The assessment and resultant extent of fire resistance can then be replicated for all remaining compartments representing a concern of external fire spread.

The assessment follows the guidance of BRE Guide BR 187. The calculations and figures below detail the results.

Following BR 187, each compartment façade relates to an enclosing compartment (multiple of 3m) with a required distance to the relevant boundary to provide 100% acceptable unprotected allowance of the compartment façade.

Consideration is also given to external fire spread via adjacent unprotected openings where the facades abut to another accommodation. In this case it is recommended that no unprotected openings between independent adjacent accommodations (forming an angle equal to or less than 80°) are closer than 1.8m.

Unprotected openings of <0.1m² in area are permitted providing even spacing of no less than 1.5m apart from one another. Larger openings of <1.0m² are also permitted providing no further opening of such size (per compartment) is located within 4.0m.

From information available we have determined the following: -

- Boundary distance - Distance from a building elevation to the boundary
- Fire compartment - A space in a building bound by fire-resisting barriers
- Separation distance - Distance from a building elevation to an adjacent building
- Unprotected area - Area of a building elevation considered for calculating boundary distance

Process

The process of determining the minimum boundary distance for a building has many steps that are common to all the available calculation methods:

- Identifying a plane of reference for each elevation from where the boundary distance needs to be calculated.
- Determining the 'unprotected areas' on each elevation. This includes any combustible cladding. As this elevation is mainly concrete structure with glazing no cladding calculations are required.
- Identification of internal compartmentation that can be used to subdivide areas on an elevation.
- The compartment with the largest amount of unprotected areas on an elevation would normally be the only area that would need to be considered.
- On this basis each floor level is considered a fire compartment from the other floor levels therefore the largest exposed area [ground floor] is considered.
- Establishing the purpose group for the building so that the appropriate compartment temperature (and thereby radiation intensity at the unprotected areas) can be selected. The purpose group for this building both purpose group 4 [shop] and purpose group 5 assembly and recreation have been used.
- The presence or not of a suppression system.
- Calculate the minimum boundary distance from each elevation using the 'enclosing rectangles' method. This may require refinement if there are local concentrations of

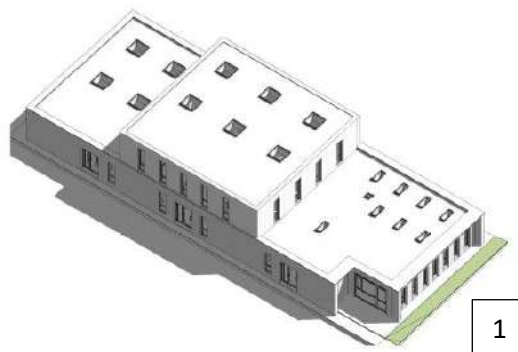
unprotected areas that would cause 'hot spots.' If the resulting minimum boundary is within the relevant boundary, then the process is complete.

For the enclosing rectangles method, the total unprotected area is found and divided by the area of a rectangle that encloses all the unprotected areas, this is expressed as a percentage. Tables A to J of BR187 can then be used to find the minimum boundary distance for the elevation.

Diagram 13



Elevation 1



On the following basis we calculate the plane of reference to the Elevation 1 to be the ground-floor level of the building.

Each floor level is considered to be a fire compartment [please see above]

The nominal boundary is calculated to the centre line of Dorset Road.

Using the appropriate table the following is calculated: [rounding up measurements]

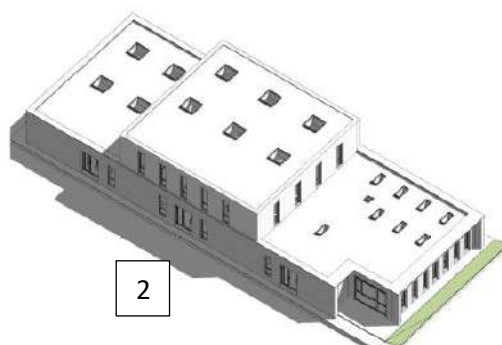
Figures in brackets are for assembly and recreation buildings.

Enclosed rectangle 6m high and the overall length of the elevation of the building is calculated at 23.5m.

Table 10

Table B – Enclosing rectangle at 6m high									
Distance from relevant boundary for unprotected percentage not exceeding									
Minimum boundary distance [m] – Figures in brackets for residential, office and assembly use									
Width	20%	30%	40%	50%	60%	70%	80%	90%	100%
3.0	1.5 [1.0]	2.0 [1.0]	2.5 [1.5]	3.0 [1.5]	3.0 [2.0]	3.5 [2.0]	3.5 [2.5]	4.0 [2.5]	4.0 [3.0]
6.0	2.0 [1.0]	3.0 [1.5]	3.5 [2.0]	4.0 [2.5]	4.5 [3.0]	5.0 [3.0]	5.5 [3.5]	5.5 [4.0]	6.0 [4.0]
9.0	2.5 [1.0]	3.5 [2.0]	4.5 [2.5]	5.0 [3.0]	5.5 [3.5]	6.0 [4.0]	6.5 [4.5]	7.0 [4.5]	7.5 [5.0]
12.0	3.0 [1.0]	4.0 [2.0]	5.0 [3.0]	5.5 [3.5]	6.0 [4.0]	7.0 [4.5]	7.5 [5.0]	8.0 [5.0]	8.5 [5.5]
15.0	3.0 [1.5]	4.5 [2.5]	5.5 [3.0]	6.0 [4.0]	7.0 [4.5]	7.5 [5.0]	8.0 [5.5]	8.5 [5.5]	9.0 [6.0]
18.0	3.5 [1.5]	4.5 [2.5]	5.5 [3.5]	6.5 [4.0]	7.5 [4.5]	8.0 [5.0]	9.0 [5.5]	9.5 [6.0]	10.0 [6.5]
21.0	3.5 [1.5]	5.0 [2.5]	6.0 [3.5]	7.0 [4.0]	8.0 [5.0]	8.5 [5.5]	9.5 [6.0]	10.0 [6.5]	10.5 [7.0]
24.0	3.5 [1.5]	5.0 [2.5]	6.5 [3.5]	7.5 [4.5]	8.5 [5.0]	9.0 [5.5]	10.0 [6.0]	10.5 [7.0]	11.5 [7.5]
27.0	3.5 [1.5]	5.0 [2.5]	6.5 [3.5]	7.5 [4.5]	8.5 [5.0]	9.5 [6.0]	10.5 [6.5]	11.0 [7.0]	12.0 [7.5]
30.0	3.5 [1.5]	5.5 [2.5]	6.5 [3.5]	8.0 [4.5]	9.0 [5.0]	10.0 [6.0]	11.0 [6.5]	11.5 [7.5]	12.5 [8.0]
40.0	3.5 [1.5]	5.5 [2.5]	7.0 [3.5]	8.5 [4.5]	10.0 [5.5]	11.0 [6.5]	12.0 [7.0]	13.0 [8.0]	14.0 [8.5]
50.0	3.5 [1.5]	5.5 [2.5]	7.5 [3.5]	9.0 [4.5]	10.5 [5.5]	11.5 [6.5]	13.0 [7.5]	14.0 [8.0]	15.0 [9.0]
60.0	3.5 [1.5]	5.5 [2.5]	7.5 [3.5]	9.0 [4.5]	11.0 [5.5]	12.0 [6.5]	13.5 [7.5]	15.0 [8.5]	16.0 [9.0]
80.0	3.5 [1.5]	6.0 [2.5]	7.5 [3.5]	9.5 [4.5]	11.5 [5.5]	13.0 [6.5]	14.5 [7.5]	16.0 [8.5]	17.0 [9.5]
100.0	3.5 [1.5]	6.0 [2.5]	8.0 [3.5]	9.5 [4.5]	11.5 [6.0]	13.5 [7.0]	15.0 [8.0]	16.5 [8.5]	18.0 [9.5]
120.0	3.5 [1.5]	6.0 [2.5]	8.0 [3.5]	10.0 [5.0]	11.5 [6.0]	13.5 [7.0]	15.5 [8.0]	17.0 [9.0]	18.5 [9.5]
130.0	3.5 [1.5]	6.0 [2.5]	8.0 [3.5]	10.0 [5.0]	11.5 [6.0]	13.5 [7.0]	15.5 [8.0]	17.0 [9.0]	19.0 [10.0]

Elevation 2



On the following basis we calculate the plane of reference to the Elevation 2 to be the ground-floor level of the building.

Each floor level is considered to be a fire compartment [please see above]

The nominal boundary is calculated to the site boundary line

Using the appropriate table the following is calculated: [rounding up measurements]

Figures in brackets are for assembly and recreation buildings.

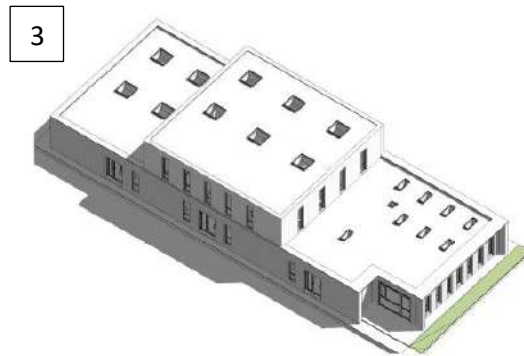
Enclosed rectangle 6m high and the overall length of the elevation of the building is calculated at 60.5m.

Table 11

Table B – Enclosing rectangle at 6m high									
Distance from relevant boundary for unprotected percentage not exceeding									
Minimum boundary distance [m] – Figures in brackets for residential, office and assembly use									
Width	20%	30%	40%	50%	60%	70%	80%	90%	100%
3.0	1.5 [1.0]	2.0 [1.0]	2.5 [1.5]	3.0 [1.5]	3.0 [2.0]	3.5 [2.0]	3.5 [2.5]	4.0 [2.5]	4.0 [3.0]
6.0	2.0 [1.0]	3.0 [1.5]	3.5 [2.0]	4.0 [2.5]	4.5 [3.0]	5.0 [3.0]	5.5 [3.5]	5.5 [4.0]	6.0 [4.0]
9.0	2.5 [1.0]	3.5 [2.0]	4.5 [2.5]	5.0 [3.0]	5.5 [3.5]	6.0 [4.0]	6.5 [4.5]	7.0 [4.5]	7.5 [5.0]
12.0	3.0 [1.0]	4.0 [2.0]	5.0 [3.0]	5.5 [3.5]	6.0 [4.0]	7.0 [4.5]	7.5 [5.0]	8.0 [5.0]	8.5 [5.5]
15.0	3.0 [1.5]	4.5 [2.5]	5.5 [3.0]	6.0 [4.0]	7.0 [4.5]	7.5 [5.0]	8.0 [5.5]	8.5 [5.5]	9.0 [6.0]
18.0	3.5 [1.5]	4.5 [2.5]	5.5 [3.5]	6.5 [4.0]	7.5 [4.5]	8.0 [5.0]	9.0 [5.5]	9.5 [6.0]	10.0 [6.5]
21.0	3.5 [1.5]	5.0 [2.5]	6.0 [3.5]	7.0 [4.0]	8.0 [5.0]	8.5 [5.5]	9.5 [6.0]	10.0 [6.5]	10.5 [7.0]
24.0	3.5 [1.5]	5.0 [2.5]	6.5 [3.5]	7.5 [4.5]	8.5 [5.0]	9.0 [5.5]	10.0 [6.0]	10.5 [7.0]	11.5 [7.5]
27.0	3.5 [1.5]	5.0 [2.5]	6.5 [3.5]	7.5 [4.5]	8.5 [5.0]	9.5 [6.0]	10.5 [6.5]	11.0 [7.0]	12.0 [7.5]

30.0	3.5 [1.5]	5.5 [2.5]	6.5 [3.5]	8.0 [4.5]	9.0 [5.0]	10.0 [6.0]	11.0 [6.5]	11.5 [7.5]	12.5 [8.0]
40.0	3.5 [1.5]	5.5 [2.5]	7.0 [3.5]	8.5 [4.5]	10.0 [5.5]	11.0 [6.5]	12.0 [7.0]	13.0 [8.0]	14.0 [8.5]
50.0	3.5 [1.5]	5.5 [2.5]	7.5 [3.5]	9.0 [4.5]	10.5 [5.5]	11.5 [6.5]	13.0 [7.5]	14.0 [8.0]	15.0 [9.0]
60.0	3.5 [1.5]	5.5 [2.5]	7.5 [3.5]	9.0 [4.5]	11.0 [5.5]	12.0 [6.5]	13.5 [7.5]	15.0 [8.5]	16.0 [9.0]
80.0	3.5 [1.5]	6.0 [2.5]	7.5 [3.5]	9.5 [4.5]	11.5 [5.5]	13.0 [6.5]	14.5 [7.5]	16.0 [8.5]	17.0 [9.5]
100.0	3.5 [1.5]	6.0 [2.5]	8.0 [3.5]	9.5 [4.5]	11.5 [6.0]	13.5 [7.0]	15.0 [8.0]	16.5 [8.5]	18.0 [9.5]
120.0	3.5 [1.5]	6.0 [2.5]	8.0 [3.5]	10.0 [5.0]	11.5 [6.0]	13.5 [7.0]	15.5 [8.0]	17.0 [9.0]	18.5 [9.5]
130.0	3.5 [1.5]	6.0 [2.5]	8.0 [3.5]	10.0 [5.0]	11.5 [6.0]	13.5 [7.0]	15.5 [8.0]	17.0 [9.0]	19.0 [10.0]

Elevation 3



On the following basis we calculate the plane of reference to the Elevation 3 to be the ground-floor level of the building.

Each floor level is considered to be a fire compartment [please see above]

The nominal boundary is calculated to the site boundary line

Using the appropriate table the following is calculated: [rounding up measurements]

Figures in brackets are for assembly and recreation buildings.

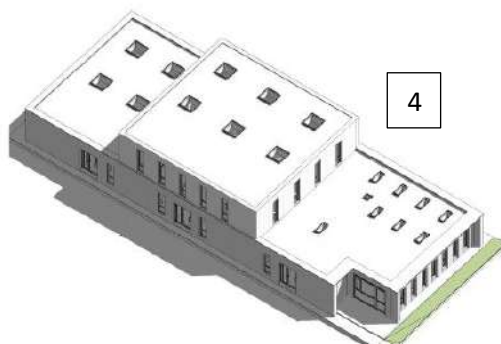
Enclosed rectangle 6m high and the overall length of the elevation of the building is calculated at 23.5m.

Table 12

Table B – Enclosing rectangle at 6m high									
Distance from relevant boundary for unprotected percentage not exceeding									
Minimum boundary distance [m] – Figures in brackets for residential, office and assembly use									
Width	20%	30%	40%	50%	60%	70%	80%	90%	100%

3.0	1.5 [1.0]	2.0 [1.0]	2.5 [1.5]	3.0 [1.5]	3.0 [2.0]	3.5 [2.0]	3.5 [2.5]	4.0 [2.5]	4.0 [3.0]
6.0	2.0 [1.0]	3.0 [1.5]	3.5 [2.0]	4.0 [2.5]	4.5 [3.0]	5.0 [3.0]	5.5 [3.5]	5.5 [4.0]	6.0 [4.0]
9.0	2.5 [1.0]	3.5 [2.0]	4.5 [2.5]	5.0 [3.0]	5.5 [3.5]	6.0 [4.0]	6.5 [4.5]	7.0 [4.5]	7.5 [5.0]
12.0	3.0 [1.0]	4.0 [2.0]	5.0 [3.0]	5.5 [3.5]	6.0 [4.0]	7.0 [4.5]	7.5 [5.0]	8.0 [5.0]	8.5 [5.5]
15.0	3.0 [1.5]	4.5 [2.5]	5.5 [3.0]	6.0 [4.0]	7.0 [4.5]	7.5 [5.0]	8.0 [5.5]	8.5 [5.5]	9.0 [6.0]
18.0	3.5 [1.5]	4.5 [2.5]	5.5 [3.5]	6.5 [4.0]	7.5 [4.5]	8.0 [5.0]	9.0 [5.5]	9.5 [6.0]	10.0 [6.5]
21.0	3.5 [1.5]	5.0 [2.5]	6.0 [3.5]	7.0 [4.0]	8.0 [5.0]	8.5 [5.5]	9.5 [6.0]	10.0 [6.5]	10.5 [7.0]
24.0	3.5 [1.5]	5.0 [2.5]	6.5 [3.5]	7.5 [4.5]	8.5 [5.0]	9.0 [5.5]	10.0 [6.0]	10.5 [7.0]	11.5 [7.5]
27.0	3.5 [1.5]	5.0 [2.5]	6.5 [3.5]	7.5 [4.5]	8.5 [5.0]	9.5 [6.0]	10.5 [6.5]	11.0 [7.0]	12.0 [7.5]
30.0	3.5 [1.5]	5.5 [2.5]	6.5 [3.5]	8.0 [4.5]	9.0 [5.0]	10.0 [6.0]	11.0 [6.5]	11.5 [7.5]	12.5 [8.0]
40.0	3.5 [1.5]	5.5 [2.5]	7.0 [3.5]	8.5 [4.5]	10.0 [5.5]	11.0 [6.5]	12.0 [7.0]	13.0 [8.0]	14.0 [8.5]
50.0	3.5 [1.5]	5.5 [2.5]	7.5 [3.5]	9.0 [4.5]	10.5 [5.5]	11.5 [6.5]	13.0 [7.5]	14.0 [8.0]	15.0 [9.0]
60.0	3.5 [1.5]	5.5 [2.5]	7.5 [3.5]	9.0 [4.5]	11.0 [5.5]	12.0 [6.5]	13.5 [7.5]	15.0 [8.5]	16.0 [9.0]
80.0	3.5 [1.5]	6.0 [2.5]	7.5 [3.5]	9.5 [4.5]	11.5 [5.5]	13.0 [6.5]	14.5 [7.5]	16.0 [8.5]	17.0 [9.5]
100.0	3.5 [1.5]	6.0 [2.5]	8.0 [3.5]	9.5 [4.5]	11.5 [6.0]	13.5 [7.0]	15.0 [8.0]	16.5 [8.5]	18.0 [9.5]
120.0	3.5 [1.5]	6.0 [2.5]	8.0 [3.5]	10.0 [5.0]	11.5 [6.0]	13.5 [7.0]	15.5 [8.0]	17.0 [9.0]	18.5 [9.5]
130.0	3.5 [1.5]	6.0 [2.5]	8.0 [3.5]	10.0 [5.0]	11.5 [6.0]	13.5 [7.0]	15.5 [8.0]	17.0 [9.0]	19.0 [10.0]

Elevation 4



On the following basis we calculate the plane of reference to the Elevation 2 to be the ground-floor level of the building.

Each floor level is considered to be a fire compartment [please see above]

The nominal boundary is calculated to the site boundary line

Using the appropriate table the following is calculated: [rounding up measurements]

Figures in brackets are for assembly and recreation buildings.

Enclosed rectangle 6m high and the overall length of the elevation of the building is calculated at 60.5m.

Table 13

Table B – Enclosing rectangle at 6m high									
Distance from relevant boundary for unprotected percentage not exceeding									
Minimum boundary distance [m] – Figures in brackets for residential, office and assembly use									
Width	20%	30%	40%	50%	60%	70%	80%	90%	100%
3.0	1.5 [1.0]	2.0 [1.0]	2.5 [1.5]	3.0 [1.5]	3.0 [2.0]	3.5 [2.0]	3.5 [2.5]	4.0 [2.5]	4.0 [3.0]
6.0	2.0 [1.0]	3.0 [1.5]	3.5 [2.0]	4.0 [2.5]	4.5 [3.0]	5.0 [3.0]	5.5 [3.5]	5.5 [4.0]	6.0 [4.0]
9.0	2.5 [1.0]	3.5 [2.0]	4.5 [2.5]	5.0 [3.0]	5.5 [3.5]	6.0 [4.0]	6.5 [4.5]	7.0 [4.5]	7.5 [5.0]
12.0	3.0 [1.0]	4.0 [2.0]	5.0 [3.0]	5.5 [3.5]	6.0 [4.0]	7.0 [4.5]	7.5 [5.0]	8.0 [5.0]	8.5 [5.5]
15.0	3.0 [1.5]	4.5 [2.5]	5.5 [3.0]	6.0 [4.0]	7.0 [4.5]	7.5 [5.0]	8.0 [5.5]	8.5 [5.5]	9.0 [6.0]
18.0	3.5 [1.5]	4.5 [2.5]	5.5 [3.5]	6.5 [4.0]	7.5 [4.5]	8.0 [5.0]	9.0 [5.5]	9.5 [6.0]	10.0 [6.5]
21.0	3.5 [1.5]	5.0 [2.5]	6.0 [3.5]	7.0 [4.0]	8.0 [5.0]	8.5 [5.5]	9.5 [6.0]	10.0 [6.5]	10.5 [7.0]
24.0	3.5 [1.5]	5.0 [2.5]	6.5 [3.5]	7.5 [4.5]	8.5 [5.0]	9.0 [5.5]	10.0 [6.0]	10.5 [7.0]	11.5 [7.5]
27.0	3.5 [1.5]	5.0 [2.5]	6.5 [3.5]	7.5 [4.5]	8.5 [5.0]	9.5 [6.0]	10.5 [6.5]	11.0 [7.0]	12.0 [7.5]
30.0	3.5 [1.5]	5.5 [2.5]	6.5 [3.5]	8.0 [4.5]	9.0 [5.0]	10.0 [6.0]	11.0 [6.5]	11.5 [7.5]	12.5 [8.0]
40.0	3.5 [1.5]	5.5 [2.5]	7.0 [3.5]	8.5 [4.5]	10.0 [5.5]	11.0 [6.5]	12.0 [7.0]	13.0 [8.0]	14.0 [8.5]
50.0	3.5 [1.5]	5.5 [2.5]	7.5 [3.5]	9.0 [4.5]	10.5 [5.5]	11.5 [6.5]	13.0 [7.5]	14.0 [8.0]	15.0 [9.0]
60.0	3.5 [1.5]	5.5 [2.5]	7.5 [3.5]	9.0 [4.5]	11.0 [5.5]	12.0 [6.5]	13.5 [7.5]	15.0 [8.5]	16.0 [9.0]
80.0	3.5 [1.5]	6.0 [2.5]	7.5 [3.5]	9.5 [4.5]	11.5 [5.5]	13.0 [6.5]	14.5 [7.5]	16.0 [8.5]	17.0 [9.5]
100.0	3.5 [1.5]	6.0 [2.5]	8.0 [3.5]	9.5 [4.5]	11.5 [6.0]	13.5 [7.0]	15.0 [8.0]	16.5 [8.5]	18.0 [9.5]
120.0	3.5 [1.5]	6.0 [2.5]	8.0 [3.5]	10.0 [5.0]	11.5 [6.0]	13.5 [7.0]	15.5 [8.0]	17.0 [9.0]	18.5 [9.5]
130.0	3.5 [1.5]	6.0 [2.5]	8.0 [3.5]	10.0 [5.0]	11.5 [6.0]	13.5 [7.0]	15.5 [8.0]	17.0 [9.0]	19.0 [10.0]

Conclusion

All adjacency distances from other buildings are compliant with the above measurements.

10.4 External Fire Spread Classification

The final external surface of the external walls should be of a spread of flame classification to adhere to Diagram 40 of Approved Document B represented in the following table.

Table 14 External Wall Classification

Building height¹	Boundary distance	Minimum external wall classification
<18m	>1.0m	No restriction
<18m	<1.0m	Class 0 (National Class) / Class B-s3, d2 (European Class)
≥18m	>1.0m	<18m - Index (I) not more than 20 (National Class) or Class C-s3, d2 (European Class)
≥18m	>1.0m	>18m - Class 0 (National Class) / Class B-s3, d2 (European Class)
≥18m	<1.0m	Class 0 (National Class) / Class B-s3, d2 (European Class)

¹Measured from ground to mean roof level as per Diagram C4 of Approved Document B

10.5 Roof Coverings

The final external surface of the proposed roof coverings should be of a spread of flame classification to impede the spread of fire along the roof between compartments (where a compartment line meets the roof junction) and to reduce the spread of fire to roofs from exposure to fire from the outside.

10.6 Roof classification at junction of compartment wall to roof

Roof structures do not require fire resistance but can represent a risk of fire spread from one compartment to another at the junction of a compartment wall to a roof membrane.

The fire spread classification of the roof material will be limited to a specified designation dependent on the minimum distance from any point of the roof to the proximate relevant boundary as per the table below.

Table 15 Roof Classification

Designation of roof covering		Minimum distance to relevant boundary			
National Class	European Class	<6m	6m-12m	12m-20m	>20m
AA, AB or AC1	BROOF(t4)	Acceptable			
BA, BB or BC	CROOF(t4)	Unacceptable	Acceptable		
CA, CB or CC	DROOF(t4)	Unacceptable	Acceptable		
AD, BD or CD	EROOF(t4)	Unacceptable			
DA, DB, DC or DD	FROOF(t4)	Unacceptable			Acceptable

¹Unwired glazing at least 4mm thick can be regarded as AA / BRoof(t4) designation

11.0 B5. Fire Service Requirements.

11.1 Fire Service Access General Principles - Existing Arrangements

To assist with the purpose of protecting life via Fire and Rescue Services operations, the building should be provided with a specific strategy for access and supporting facilities to assist with the attending Fire Service actions.

Schedule 1 of the Building Regulations requires the following functional requirement to be met in respect of B5, Access and Facilities for the Fire Service:

- (1) The building shall be designed and constructed so as to provide reasonable facilities to assist fire fighters in the protection of life.
- (2) Reasonable provisions shall be made within the site of the building to enable fire appliances to gain access to the building.

The following discusses the implications of these requirements on the proposed design with regard to access and facilities for the Fire Service within and around the building(s).

Dependent on the building use and size, this involves providing the following:

- Adequate means of additional water supply,
- Adequate means for vehicular access to the development (externally),
- Adequate means for firefighting personnel access to the development (internally),
- Supporting provisions for internal firefighting such as fire mains and firefighting shafts (if any), and
- Ventilation to sub-ground floor areas (if required).

11.2 Hydrant and Water Supply Access

Approved Document B Section 15 recommends that public hydrants should be present to ensure that rising main inlets or the building(s) entrances are accessible within 90m of an available hydrant with further hydrants no more than 90m apart from one another.

If the above is unachievable, further private hydrant(s) should be included on site where necessary or alternative arrangements made through liaison the Local Fire Authority's water services team.

The final hydrant access strategy shall remain as existing arrangements which are considered adequate and sufficient.

Fire hydrants are provided within Dorset Road and are sited under 90m away from the building [Existing Condition].

11.3 External Fire Service Vehicle Access

Fire Service vehicle access is available to the site via both Brookhill Road.

Table 16 Typical Fire & Rescue Service Vehicle Access Route Specification – Existing arrangements

Appliance type¹	Minimum width of road between kerbs	Minimum width of gateways	Minimum turning circle between kerbs	Minimum turning circle between walls	Minimum clearance height	Minimum carrying capacity
<i>Pump</i>	3.7m	3.1m	16.8m	19.2m	3.7m	14.0t
<i>High reach</i>	3.7m	3.1m	26.0m	29.0m	4.0m	23.0t

¹Non-standardised appliance types may be applicable, dependent on Local Fire Authority.

Any access / security measures in and around the site (especially any bollards preventing vehicle access) will need to be by-passable by the Fire Service. The details of the bypass arrangements will need to be developed and agreed with the local Fire Service as applicable.

Where dead-end routes are evident within the site, these will either be limited to approximately 20m or a turnaround (turning circle or hammerhead) facility provided.

To consider the relationship between the building perimeter and hardstanding / access road for the high reach appliance, the following should be achieved.

Table 17 Design Requirement for High Reach Appliances – Existing arrangements

Design requirement	Dimension¹	Comment
<i>Maximum distance of building external wall to near edge of access route</i>	2.0m	<i>Overhead obstruction / roof overhang to be avoided</i>
<i>Minimum width of access route</i>	5.5m	<i>To allow for manoeuvring</i>
<i>Minimum distance between furthest edge of access route and obstruction</i>	2.2m	<i>For swing of appliance platform</i>

¹Non-standardised appliance types may be applicable, dependent on Local Fire Authority.

Diagram 14



11.4 General Firefighting Provisions

Information boxes contain relevant fire strategy plans and current information will be located at the main entrance to the community centre with prayer facilities building.

12.0 Fire Safety Systems

12.1 Portable Fire Fighting Equipment

Portable firefighting equipment provided, must comply with BS:5306 and tested and maintained in accordance with this standard.

Typically, firefighting media [fire extinguishers in this case] must be positioned in the following areas.

- Main entrance / exit
- Fire exit / escape doors
- Along fire exit route

A person should not.

- Travel more than 25m to reach access to a fire extinguisher.
- Go through more than two doors to reach access to a fire extinguisher.
- Go to a different level or floor to reach access to a fire extinguisher [not applicable in this instance]

Management must ensure that monthly visual inspections of the fire extinguishers are undertaken and recorded in the fire safety logbook.

Inspections should include the following.

- The fire extinguishers are located in the designated positions.
- They are unobstructed, visible and its operating instructions face outwards.
- Has operating instructions which are clean and legible?
- Are not obviously damaged.
- Has a reading in the operable range [Green]?
- Has seals and tamper indicators which are not broken or missing.
- Are provided with signage indicating the type of fire extinguisher which should be located at the fire extinguisher point.
- All records and certification relating to testing and maintenance of the fire extinguishers must be recorded and made available for clarification upon request.

Note: BS5306-3 [Section 5.4.3 – Use of powder extinguishers] advises that the discharge of a dry powder extinguisher within confined spaces can cause a sudden reduction of visibility and can also impair breathing which could jeopardise escape, rescue, or any other emergency action. For this reason, personal safety must be considered where the use of such an extinguisher may be a requirement.

Table 19 Fire Extinguisher Classification

Type	Colour	Class A General combustible materials (wood, paper etc.,)	Class B Flammable liquid material (paint, petrol etc.,)	Class C Flammable gases (methane, butane etc.,)	Class D Flammable metals (sodium, lithium, potassium etc.,)	Class E Electrical equipment (computers etc.,)	Class F Deep fat fryers (chip pans or cooking fats / oils etc.,)
Water		acceptable	Not acceptable	Not acceptable	Not acceptable	Not acceptable	Not acceptable
Foam		acceptable	acceptable	Not acceptable	Not acceptable	Not acceptable	Not acceptable
Dry Powder		Not acceptable	Not acceptable	Not acceptable	Not acceptable	Not acceptable	Not acceptable
Carbon Dioxide		acceptable	acceptable	Not acceptable	Not acceptable	acceptable	Not acceptable
Wet Chemical		acceptable	acceptable	Not acceptable	Not acceptable	Not acceptable	acceptable

**Note dry powder is considered not acceptable in a residential environment and synthetic foam extinguishers are due to be phased out by 2026.*

12.2 Escape Route Signage

Fire exit signage, either mounted on door leaves, walls, or hung at strategic positions, clearly and unambiguously indicates escape routes within and from the building.

All new, replacement or additional signage should be sized according to the predicted viewing distance and sited conspicuously in accordance with British Standard recommendations. All signs must meet the recommendations of BS 5499-1:2000 Safety Signs, including fire safety signs Part 1: Specification.

Fire escape signs are provided to guide you from wherever you are in a building, via a place of relative safety [the escape route] to the place of ultimate safety [the assembly area].

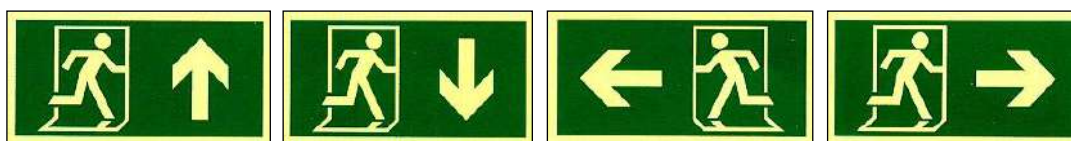
Fire escape signs are not needed on the main route into or out of a building [the one used by people for normal arrival and exit]. The alternative escape routes and complicated escape routes do need to be signed. It must not be assumed that everyone will know all safe routes through the building. Similarly, it must not be assumed that, once outside the building via a final exit, people will know how to get to the assembly area, so signs directing people to the assembly area will be needed as well.



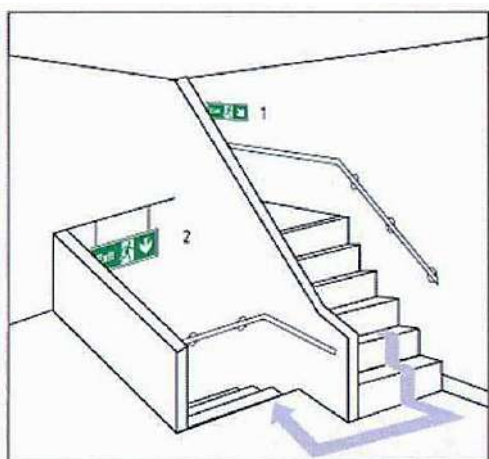
Fire escape signs are green and white - safe condition. They must comprise a pictogram, an arrow, and possibly words. A sign with just an arrow, or just words, or an arrow and words is not sufficient. There must be a pictogram including the “rapidly walking man.” Signs to British Standard 5499: part 4:2000 have the man passing through a door. Another way to identify signs to BS5499 is that the text is in lower case apart from the first letter. Signs that are illuminated from behind or within [as in the case of an escape light with escape sign incorporated] will have a white figure entering a green door with a white surround.

You may encounter signs to European Standard 92/58/EEC. These are identified by all upper-case text, a man, a door, and an arrow.

Examples of Fire Exit Signs



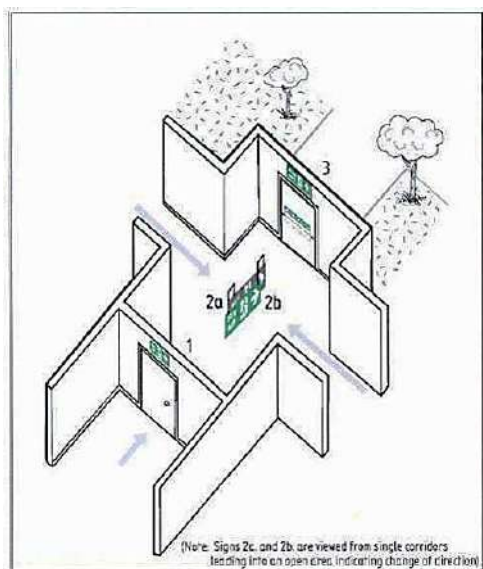
**It is recommended that photo-luminescent signs be used, they glow in the dark.
Signing the ideal Escape Route to BS5499**



Example 1:

Sign 1 means progress down to the right as viewed from in front of the sign. Sign sited on wall of half landing.

Sign 2. Progress down from here as viewed from in front of the sign. Suspended from the ceiling



Example 2:

Sign 1: Progress forward and through from here as viewed from in front of the sign. Positioned above door.

Signs 2a and 2b: Progress to the left/right from here.

Suspended at change of direction. Sign 3: Progress forward and through from here as viewed from in front of the sign. Positioned above door.

NOTE: Outside the final exit [sign 3 above it] if the door can be obstructed a “Fire Exit Keep Clear” sign is needed on the outside of the door.



Example 3:

Sign 1: Progress forward and through from here as viewed from in front of the sign. Positioned above door.

Sign 2: Progress down to the left from here as viewed from in front of the sign. Positioned on the landing

Fire escape route signage must be clear and unobstructed as well as visible from all working areas and along escape route corridors.

Diagram 15 Escape Route Signage

Escape Route Sign	Meaning	Location Example
	Progress down to the right (change of level)	On wall / suspended at head of stairs On half landing wall or stairs Suspended at change of level
	Progress up to the right (change of level) / Progress forward and across to the right from here when suspended in open area	On wall / suspended at foot of stairs On half landing wall / stairs Suspended at change of level Suspended in open areas
	Progress down to the left (change of level)	On wall / suspended at head of stairs On half landing wall / stairs Suspended at change of level
	Progress up to the left (change of level) / Progress forward and across to the left from here when suspended in open area	On wall / suspended at foot of stairs On half landing wall / stairs Suspended at change of level Suspended in open areas
	Progress forward from here / Progress forward and through (when sited above door) / Progress forward and up from here (change of level)	Suspended in corridor leading to door Suspended in open areas Suspended in front of door Positioned above door Suspended at foot of stairs or ramp
	Progress to the right from here	On corridor walls Suspended adjacent and left of the exit Suspended at change of direction
	Progress to the left from here	On corridor walls Suspended adjacent and right of the exit Suspended at change of direction
	Progress down from here (change of level)	Suspended at head of stairs / ramp Suspended at change of level

All allocated fire resisting doors which form part of the compartment boundary or fire escape route must have Fire Door signs affixed to both sides of the door leaf to ensure that there is

no confusion over its status and importance as an integral part of the compartmentation within the building.

Diagram 16 Fire Door Signage

Signage	Sign diameter / letter height	Position of signage	Location	Method of closure
	60mm sign diameter and 5mm letter height	Located on both faces of the door at eye level	Fire doors in staircases, and protected lobbies	Approved self-closing device fitted
	60mm sign diameter and 5mm letter height	Located on the front face of the door at eye level	On the outside door of the riser Cupboards, maintenance access Cupboards, Storerooms, Utility rooms, and cleaners Cupboards	Not required
	60mm sign diameter and 5mm letter height	Located on the open face of the door at eye level	Doors nominated as Automatic Opening Vents / doors Cross-passage doors	Approved self-closing device fitted

Fire resisting doors fitted to hazard rooms or cupboards which have not been fitted with self-closing devices should remain locked and have the appropriate signage affixed to the external side of the door leaf.

Fire resisting doors which have been fitted with electro-magnetic, wireless, or battery-operated door holding devices should be furnished with “Automatic Fire Door Keep Clear” signage.

Diagram 17 Fire Main Signage

Device	Signage	Sign height / letter height	Location
Dry riser main		120mm sign height and 15mm letter height	Dry rising fire main inlet and outlet
Premise information box		120mm sign height and 10mm letter height	Premise information box

Signage indicating potential hazards, for example, hazardous substances, liquids, electrical hazards etc. should be provided at the location of such areas where required giving indication of their presence.



Signage should also be in place to indicate the location of isolation points for services such as gas, electric etc. no smoking areas as well as information relating to emergency procedures.



No smoking signs should be positioned where applicable. Fire action notices must be completed and give indication of the location of the fire assembly point as well as any other relevant information applicable to such signage. It is recommended that fire action notices are provided at all call points for ease of reference.

Final exit doors should have “Fire Exit Keep Clear” affixed to the external side of the door leaf.

A fire alarm zone plan should be positioned adjacent to any fire alarm panel for ease of reference.

All fire extinguisher points should be furnished with signage indicating the type of extinguisher required to be in position at the fire point.

13.0 Utility Services

All utility services are to be installed and maintained according to current regulations.

14.0 Regulation 38

Intention

The aim of this regulation is to ensure that the person responsible for the building has sufficient information relating to fire safety to enable them to manage the building effectively. The aim of regulation 38 will be achieved when the person responsible for the building has all the information to enable them to do all of the following: -

- Understand and implement the fire safety strategy of the building.
- Maintain any fire safety system provided in the building.
- Carry out an effective fire risk assessment of the building.

Fire safety information

For building work involving the erection or extension of a relevant building (i.e. a building to which the Regulatory Reform (Fire Safety) Order 2005 applies or will apply), or the relevant change of use of a building, fire safety information should be given to the responsible person at one of the following times.

- When the project is complete.
- When the building or extension is first occupied.

This section is a guide to the information that should be provided. Guidance is in terms of essential information and additional information for complex buildings; however, the level of detail required should be considered on a case-by-case basis.

Essential information

Basic information on the location of fire protection measures may be sufficient. An as-built plan of the building should be provided showing all of the following.

- a. Escape routes – this should include exit capacity (i.e. the maximum allowable number of people for each storey and for the building).
- b. Location of fire-separating elements (including cavity barriers in walk-in spaces).
- c. Fire door-sets, fire door-sets fitted with a self-closing device and other doors equipped with relevant hardware.
- d. Locations of fire and/or smoke detector heads, alarm call points, detection/alarm control boxes, alarm sounders, fire safety signage, emergency lighting, fire extinguishers, dry or wet fire mains and other firefighting equipment, and hydrants outside the building.
- e. Any sprinkler systems, including isolating valves and control equipment.
- f. Any smoke control systems, or ventilation systems with a smoke control function, including mode of operation and control systems.
- g. Any high-risk areas (e.g. heating machinery).

Details should be provided of all of the following.

- a. Specifications of any fire safety equipment provided, including routine maintenance schedules.
- b. Any assumptions regarding the management of the building in the design of the fire safety arrangements.
- c. Any provision enabling the evacuation of disabled people, which can be used when designing suitable personal emergency evacuation plans

Additional information for complex buildings

A detailed record should be provided of both of the following.

- a. The fire safety strategy.
- b. Procedures for operating and maintaining any fire protection measures. This should include an outline cause and effect matrix/strategy for the building. Further guidance is available in clause 9 and Annex H of BS 9999.

The records should include details of all of the following.

- a. The fire safety strategy, including all assumptions in the design of the fire safety systems (such as fire load). Any risk assessments or risk analysis.
- b. All assumptions in the design of the fire safety arrangements for the management of the building.

All of the following.

- Escape routes (including occupant load and capacity of escape routes).
- Any provision to enable the evacuation of disabled people.
- Escape strategy (e.g. simultaneous or phased).
- Muster points.
- All passive fire safety measures, including all of the following.

- Compartmentation (i.e. location of fire-separating elements).
- Cavity barriers.
- Fire door-sets, including fire door-sets fitted with a self-closing device and other doors equipped with relevant hardware (e.g. electronic security locks).
- Duct dampers.
- Fire shutters.
- Fire detector heads
- Smoke detector heads.
- Alarm call points.
- Detection/alarm control boxes.
- Alarm sounders.
- Emergency communications systems
- CCTV.
- Fire safety signage.
- Emergency lighting.
- Fire extinguishers.
- Dry or wet fire mains and other firefighting equipment
- Other interior facilities for the fire and rescue service.
- Emergency control rooms.
- Location of hydrants outside the building.
- Other exterior facilities for the fire and rescue service.
- All active fire safety measures, including both of the following.
- Sprinkler system(s) design, including isolating valves and control equipment.
- Smoke control system(s) (or heating, ventilating and air conditioning system with a smoke control function) design, including mode of operation and control systems.
- Any high-risk areas (e.g. heating machinery) and particular hazards
- Plans of the building as built, showing the locations of the above
- Both of the following.
- Specifications of any fire safety equipment provided, including all of the following.
- Operational details.
- Operators' manuals.
- Software.
- System zoning.
- Routine inspection, testing and maintenance schedules.
- Records of any acceptance or commissioning tests.
- Any other details appropriate for the specific building

15.0

Project Plans

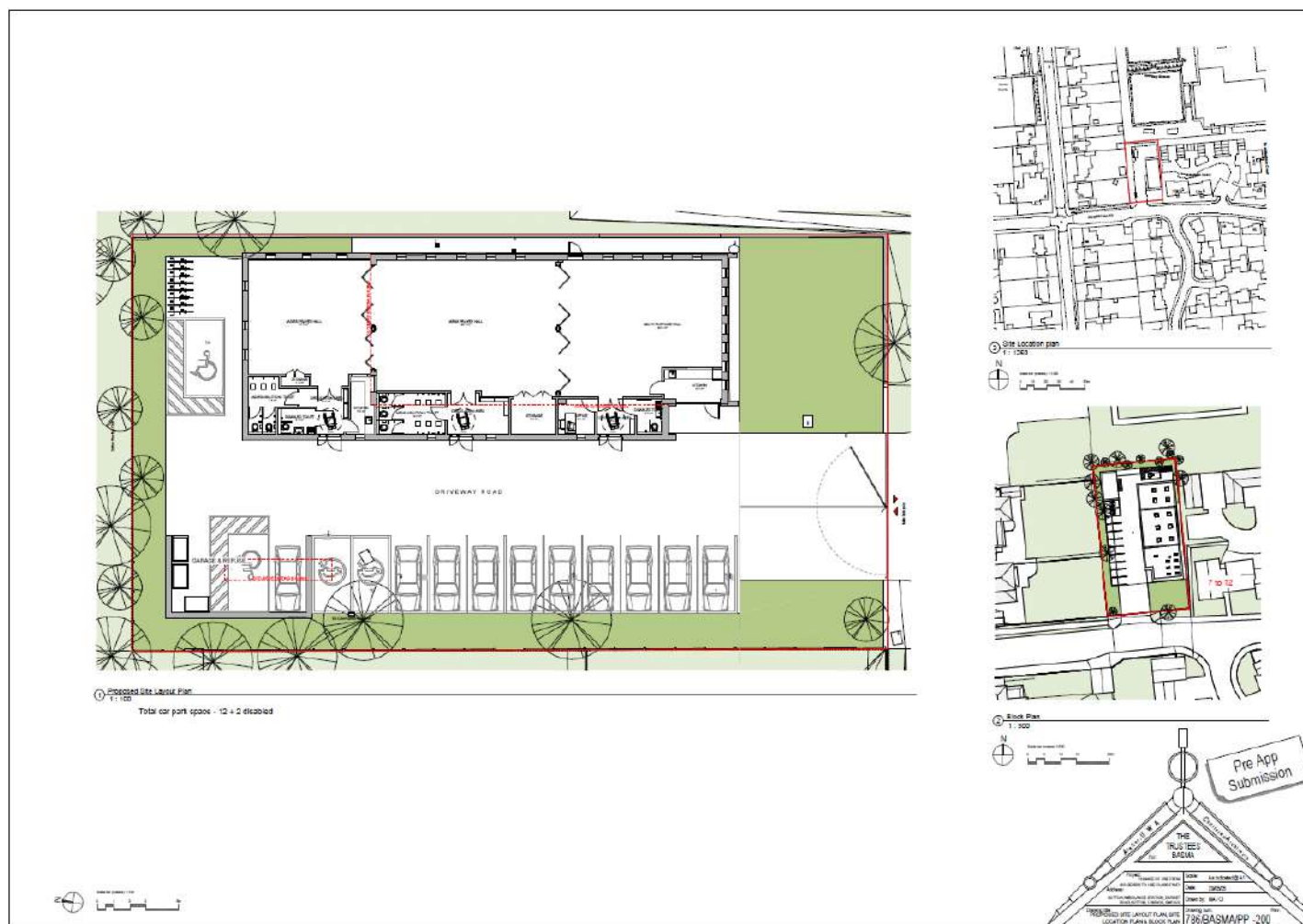


Diagram 18 786/BASMA/PP-200 — Proposed Site Layout Plan, Site Location Plan & Block Plan

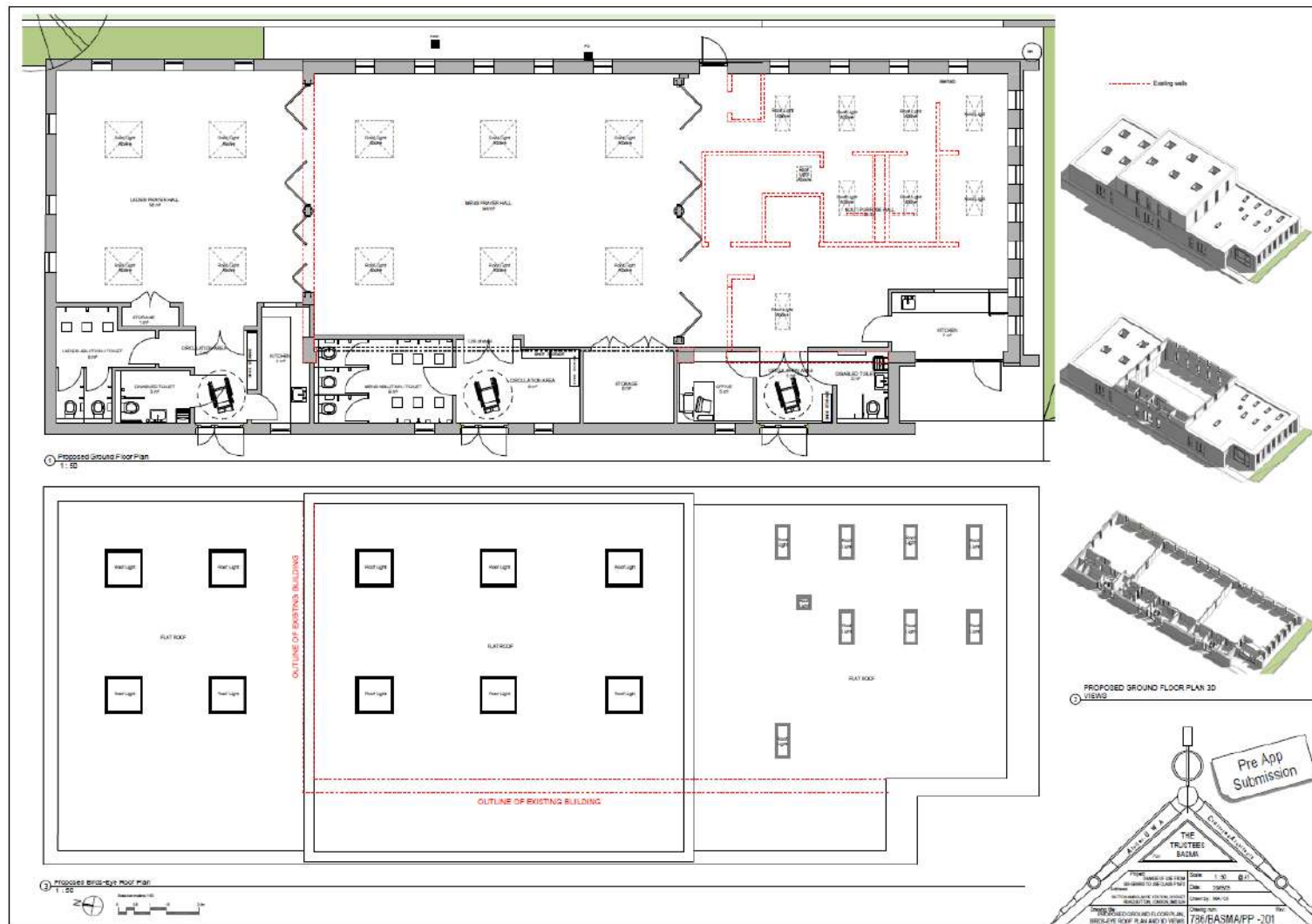


Diagram 19 786/BASMA/PP-201 — Proposed Ground Floor Plan, Birds-Eye Roof Plan and 3D View

Materials Key	
1 /	Timber Geometric Screen
2 /	Charred Timber Shiguban Cladding
3 /	Timber Slats
4 /	Patterned Red Brick
5 /	Green Climbing Plants



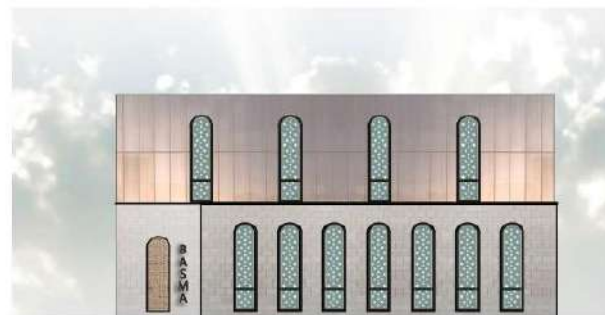
BASMA FRONT MATERIALITY ELEVATION - OPTION A



BASMA SIDE MATERIALITY ELEVATION - OPTION A

Diagram 20 786/BASMA/PP-202 — Proposed Material Elevation A

Materials Key
1 / White / Cream Stone
2 / Bronze Cladding
3 / Red and Beige Stone Detailing



BASMA FRONT MATERIALITY ELEVATION - OPTION B



BASMA SIDE MATERIALITY ELEVATION - OPTION B

Diagram 21 786/BASMA/PP-203 — Proposed Material Elevation B



Diagram 22 786/BASMA/PP-204 — Proposed Sections

16.0 Management Roles and Responsibilities

16.1 Statutory Requirements

This Building is subject to control under the Regulatory Reform (Fire Safety) Order 2005, with the Chief Executive [designated to the Director of Risk Management for the company] having the ultimate responsibility for compliance.

16.2 Duty of Care Responsibilities

The responsible person, as designated within the fire policy, will make, and give effect to such arrangements as are appropriate, having regards to the size of his/her undertaking and the nature of his/her activities, for the effective planning, organisation, control monitoring, and review of the preventive and protective measures for life safety, property protection, and duty of care, that needs to be delivered for a safe and secure operation.

The company has appointed competent persons to assist in undertaking the preventive and protective measures to ensure, as far as is reasonably practicable, the safety of any of his/her employees and, in relation to relevant persons who are not his employees, to take such general fire precautions as may reasonably be required in the circumstances of the case to ensure the premises are safe.

16.3 Responsible Person

The Responsible Person means a person who has control of the premises (as occupier or otherwise) in connection with the carrying on by him of a trade, business, or other undertaking (for profit or not).

16.4 Competent Person

A person is to be regarded as competent where they have sufficient training and experience or knowledge and other qualities to enable them properly to assist in undertaking the preventive and protective measures.

The responsible person must ensure any competent persons appointed are provided with the adequate means at their disposal, having regard to the size of the premises, the risks to which relevant persons are exposed and the distribution of those risks throughout the premises.

16.5 Fire Risk Assessment – Design and construction phase

Those responsible for the design and construction of the building will provide fire safety information to the responsible person at the completion of the project or when the building is first occupied, this will include a fire risk assessment carried out by a competent person.

It is a requirement under Article 9 of the Regulatory Reform (Fire Safety) Order 2005, that the Responsible Person carries out a suitable and sufficient fire risk assessment of the premises of the risks to which relevant persons are exposed for the purpose of identifying the general fire precautions which need to be taken to comply with the requirements and prohibitions imposed by or under the Fire Safety Order, and any significant findings are recorded.

16.6 Fire Risk Assessment – Once work has been completed

It is a requirement under Article 9 of the Regulatory Reform (Fire Safety) Order 2005, that the company carries out a suitable and sufficient fire risk assessment of the premises of the risks to which relevant persons are exposed for the purpose of identifying the general fire precautions which need to be taken to comply with the requirements and prohibitions imposed by or under the Fire Safety Order, and any significant findings are recorded.

The process of risk assessment is an ongoing task that is constantly monitored and reviewed regularly so as to keep it up to date, particularly if there is reason to suspect that it is no longer valid or there has been a significant change in the matters to which it relates, including when the premises, special, technical, and organisational measures, or organisation of the work undergo significant changes, extensions, or conversions.

A fire safety risk assessment should be carried out after three months of occupation to ensure working practices have not compromised the existing fire safety measures and therefore increased either the fire risk or life risk.

The process of risk assessment is an ongoing task that is constantly monitored and reviewed regularly so as to keep it up to date, particularly if there is reason to suspect that it is no longer valid or there has been a significant change in the matters to which it relates, including when the premises, special, technical, and organisational measures, or organisation of the work undergo significant changes, extensions, or conversions.

The Responsible Person is liable in law if the fire risk assessment is not suitable and sufficient. It is important, therefore, that you show all due diligence in ensuring that the fire risk assessor is competent.

If, as a result of an inadequate fire risk assessment, people are placed at risk of death or serious injury in the event of fire, the Responsible Person who is required by the Fire Safety Order to carry out the fire risk assessment (and the fire risk assessors themselves) have committed an offence for which they can be prosecuted.

It is also important that you have access to competent advice on compliance with fire safety legislation (preferably from someone within your organisation, but, otherwise, from external advisers, such as your fire risk assessor or even your local fire and rescue service).

A fire risk assessment is not a 'one off' exercise. There is a need to review it regularly to keep it up to date. The Fire Safety Order does not specify the frequency with which fire risk assessments should be reviewed. However, it specifies that reviews must be carried out regularly, and will in particular be required if there is reason to suspect it may no longer be valid or if there has been a significant relevant change, e.g., to the premises or the activities within them.

Your fire risk assessor should advise you as to the frequency with which reviews should be carried out. However, for small and simple premises, an annual review is recognised as good practice.

On this basis we would advise you that the building is considered to fall within the above category and a fire risk assessment should be carried out within 12 months from the anniversary of the previous assessment and must be completed by a person who is deemed competent in fire safety matters.

Fire risk assessments must be based on the following: -



16.7 Maintenance of Fire Safety Equipment and Provisions

The Responsible Person must have in place a planned inspection, maintenance, and testing programme to ensure all fire protection systems are fit for service and can operate effectively when required in an emergency situation.

Arrangements and contracts are in place with competent staff and specialist contractors for maintenance to be carried out on all life safety and fire protection systems with records retained on site. This includes alterations, additions, repairs or modifications to services and equipment.

16.8 Training

The Management Team need to be trained to a level whereby they fully understand their role and responsibilities, in line with the Fire Safety Regulatory Reform Order 2005. The subjects that need to be covered include:

- ☐ Planning and legal framework
- ☐ Emergency Planning
- ☐ Integrated Emergency Management
- ☐ The combined response of emergency services
- ☐ Contingency planning
- ☐ Business continuity
- ☐ Communications
- ☐ Development and testing of plans.
- ☐ Development of Site-Specific Emergency plans
- ☐ Command and control of operations
- ☐ Public Relations

16.9 Statutory Records

One of the essential elements that Management control is the holding of statutory records, keeping them up to date, (either in manual or electronic form), on site so that they can be made available to any of the statutory enforcing bodies on request.

These records should include:

-  Fire Alarm and Detection System
-  Fire Safety Signage
-  Firefighting Equipment
-  Emergency Escape Lighting System
-  Fire Compartmentation Doors
-  Fire Drill Training
-  Gas Installations
-  Electrical Portable Appliance Testing
-  Fixed wiring electrical installation
-  Lightning protection
-  Private Hydrants
-  Fault or modification to life safety/property protection systems

Where changes are made to the building as a result of refurbishment or new projects, these will be reviewed and updated by the Fire Safety Consultant, who will advise the responsible person accordingly.

16.10 Security Arrangements

The Responsible Person or delegated person must be trained to the required security standards and also possess the ability to carry out key special tasks, which include:

- ☐ Meeting Emergency Services
- ☐ Responding to incidents.
- ☐ Use of first aid fire extinguishers.
- ☐ Incident report production
- ☐ Fault reports

16.11 Management Fire Safety Policy

The promulgation of the company's fire safety policy and approach to life safety, property protection, and incident management planning, within the company has been prepared in accordance with the guidelines of the Regulatory Reform Order 2005

The primary purpose of this policy is the protection of life safety for its, users, staff, visitors, and fire service personnel, by ensuring compliance with all relevant statutory controls, to ensure that measures provided for fire safety are compatible with the operational needs of the company, so far as is practicable.

This is promulgated by making all staff aware of the importance of fire safety and ensures that they receive appropriate training at different levels, commensurate with their duties, so that they can discharge their responsibilities effectively to reduce the risk from fire to the company and its contents, and ensures that the required standards of fire safety in the buildings are regularly audited and inspected in line with the fire safety policy so that it can be effectively managed to reduce the danger from fire to be as low as is reasonably practicable

16.12 Emergency Procedures Plans

There must be a comprehensive emergency procedures plan developed for the building which is reviewed and updated on a regular basis.

The key elements of the procedures include: -

- ☐ ☐ Action on discovering a fire.
- ☐ ☐ Warning and evacuation signals.
- ☐ ☐ Calling the Fire and Rescue Service
- ☐ ☐ Providing information to the Fire and Rescue Service
- ☐ ☐ Evacuation process
- ☐ ☐ First aid firefighting measures
- ☐ ☐ Assembly points
- ☐ ☐ Duties of Fire Wardens
- ☐ ☐ Assistance with special needs
- ☐ ☐ Site plans
- ☐ ☐ Re-entry into isolated areas of the building

16.13 Control of Substances Hazardous to Health (COSHH)

Hazardous substances to health have become a major importance to emergency response teams and needs to be carefully controlled and managed to mitigate the risk to users, visitors, employees, and Fire Service personnel who may have to deal with such an incident.

Regardless of its size or nature, each COSHH incident can present a potential hostile environment. While preventing exposures to hazardous substances is always the primary concern, it also important to be able to identify the location and type of substances to enable the Emergency Services to deal with the event in a safe manner.

All COSHH substances are readily identified within the area stored and are satisfactorily controlled. Copies of assessments should be held in a central file with other Building Emergency Documents so that they can be quickly made available to the Fire Service when they have to deal with an incident.

16.14 Contingency Planning

Contingency plans need to include preparation and response to a wide range of unusual events.

This can include possible emergencies and incidents that include fire, communications, water, power supplies, weather, and should take account of time of day, time of week, time of year, traffic issues, and other unexpected/unplanned events.

It is essential that when Management prepare their contingency plan that these key elements are discussed and incorporated into the final plan, as necessary.

For any plan to be successful it needs to be kept up to date by regular review and should be tested by practical or tabletop exercises involving all interested parties, including the emergency services.

16.15 Incident Management Systems Operations

The Incident Management System Operations should be considered as the basic command control process to be used to deal with any size or kind of incident that may occur.

This process has to be viewed as an extension of the emergency plan and needs to be supported by the Management Team to provide the four key elements such as.

- ☐ ☐ Strategic Level – overall direction of the incident

- Tactical Level - assigns and sets objectives for other incident team members.
- Task Level - assigns specific tasks to service partners and concessionaires.
- Operations Level – coordinate and provide information to Emergency Services.

16.16 Command Organisation

For this Management to be successful it is important to appoint a Senior Position from the Management Team as the Emergency Co-ordinator to set up Command and Control and liaise with the Fire Service.

This person should report to the Senior Management Team, have responsibility for ensuring the Control Centre is available to the Emergency Services, be properly briefed and trained and coordinate the activities.

17.0 Legislative Drivers

The main fire legislation applicable to this building includes, The Building Regulations 2010 (as amended – 2019 edition with 2020 amendments Part B Volume 1) and The Regulatory Reform (Fire Safety) Order 2005.

This document forms a concept approach for fire matters; the Design Team must ensure the contents of this report are incorporated in the building. This concept will not prevent a fire occurring and good housekeeping will be encouraged to reduce the risk. This strategy is mainly concerned with getting occupants out of the building safely and providing measures, where necessary, to assist the fire fighters in their operations.

The concept is only valid where the systems are designed correctly and maintained in an operating condition. If there is a failure in the management approach and a fire occurs, this concept may not reduce the impact on contents and building damage. Until this report is agreed with the approving authorities, the content should only be used as preliminary information.

Following occupation, the management of the premises are required under current legislation to carry out a fire risk assessment. A fire safety risk assessment will be carried out annually, or when there are material changes to either layout, or working practices, where an assessment will be carried out immediately after any such occurrence. This document will be developed following completion of the works and will form part of the fire manuals developed for the premises.

17.1 Building Regulations

The construction or modification of any building in England and Wales needs to comply with the statutory requirements of the Building Regulations. These regulations deal with the set minimum standards of design and building work for the construction of domestic, commercial, and industrial buildings. The Building Regulations contain a list of requirements, referred to in Schedule 1, which purpose is, amongst other provisions, the health and safety of people in and around buildings. There are 15 Parts, which cover subjects such as structure, fire safety, ventilation, drainage, etc.

In the case of fire, the regulations are dealt with under the functional requirements parts B1 to B5 of Schedule 1 of the Building Regulations. There are a number of guidance documents, which can be adopted to show compliance with the Schedules. These include

ADB and other British Standards. These guides are considered as adequate to provide general guidance for the more common buildings.

An alternative approach is to adopt Fire Safety Engineering, which integrates fire engineering calculations, life safety systems, building inherent features and professional judgement, to produce a Fire Strategy that achieves appropriate levels of safety to a specific building and use.

These include BS 9991 and BS 9999, which are considered as adequate to provide general guidance for the *more* common approaches in building design and where scenarios move away from traditional approaches from Approved Document B.

- B1 - Means of Warning and Escape
- B2 - Internal Fire Spread (Linings)
- B3 - Internal Fire Spread (Structure)
- B4 - External Fire Spread
- B5 - Access and Facilities for the Fire Service

17.2 Regulatory Reform (Fire Safety) Order 2005

A full risk assessment covering general health and safety as well as fire safety will be required on occupation of the building in order to meet the Regulatory Reform Order and will allow the ongoing control to be monitored to ensure safe escape can be achieved. The assessment should be reviewed at regular intervals or when significant change to the building occurs.

Article 3	Defines the responsible person as a.) the employer, bi.) the person in control of the premises, or b.ii) the owner/landlord.
Article 4	Defines the meaning of general fire precautions
Article 5	States that the responsible person has an obligation to comply with the duties imposed by the order and any other person who has to any extent control of the premises, has an obligation in relation to maintenance, repair, or safety of the premises.
Article 6	Lists exemptions, which includes domestic premises occupied as a single private dwelling.
Article 8	Requires general fire precautions are undertaken to ensure the safety of employees / relevant persons.
Article 9	Requires the responsible person to carry out a fire risk assessment. This assessment must be reviewed regularly and particularly if there is reason to suspect that it is no longer valid or there has been a significant change.
Article 11	Requires the responsible person to manage the required preventive and protective arrangements and record the arrangements where a.) five or more people are employed, b.) a licence under an enactment is in force, or c.) an alterations notice is in force.
Article 12	Requires the responsible person to eliminate or reduce risks from dangerous substances.
Article 13	Requires suitable firefighting equipment and fire detection and alarm systems are provided.
Article 14	Requires that escape routes and exits are kept clear and accessible at all times.
Article 15	Requires appropriate procedures to be established including the need for fire drills and competent persons to assist in their undertaking.
Article 17	Requires suitable maintenance of fire safety facilities and equipment.
Article 19	Requires communication with employees, including the significant findings of the fire risk assessment.
Article 20	Requires that any external employers / employees who are working (or have staff working) in premises are provided with relevant information on the risks and the preventive and protective measures in place.
Article 21	Requires that employees are provided with suitable and appropriate training.
Article 22	Requires co-operation / coordination where two or more responsible persons share duties in premises.
Article 23	Lists the general duties of employees at work.
Article 24	Allows the introduction of additional regulations.

17.3 Fire Safety Act 2021

In the case of multi-occupied residential buildings, the Fire Safety Act puts beyond doubt that structure, external walls, and flat entrance doors fall within the scope of the Fire Safety Order. The Fire Safety Act will require responsible persons to ensure that these elements are included in their fire risk assessments, if they have not been covered already.

Fire risk assessments

A fire risk assessment is required by the Fire Safety Order to assess fire safety risks relevant to both workplaces and buildings with 2 or more sets of domestic premises. It is the duty of the responsible person to ensure that this is carried out, although they may appoint someone on their behalf, such as a competent fire risk assessor, to carry it out.

Legal requirements of the Fire Safety Act

Section 1 of the Fire Safety Act clarifies that where a building contains 2 or more sets of domestic premises the responsible person must take account of structure, external walls, and flat entrance doors in a fire risk assessment. The Fire Safety Act amends article 6 of the Regulatory Reform (Fire Safety) Order Act 2005 in this regard.

Section 3 of the Fire Safety Act makes clear that if a responsible person has followed the article 50 commencement guidance, which can also include use of the Fire Risk Assessment Prioritisation Tool, then they may then be able to use evidence of this compliance to demonstrate that they have met with their obligations under the Fire Safety Order (specifically the elements clarified by the Fire Safety Act). If a responsible person has not complied with the guidance or the prioritisation tool, they may need to provide alternative evidence of how they have complied with this aspect of their Fire Safety Order obligations.

Where the responsible person has already updated their fire risk assessment to include external walls

For many buildings, particularly high rise, we expect that responsible persons will already be aware of the composition of external walls and will have updated their fire risk assessment and have taken, or will take action to mitigate any risk.

Where this assessment has been carried out, for multi-occupied residential buildings of any height, the fire risk assessment does not need to be updated again other than after a review as required by article 9 of the Fire Safety Order.

Where the fire risk assessment has not already been updated to include external walls

The responsible person, or competent fire risk assessor acting on their behalf, should undertake a visual inspection of the external walls.

In many cases it will be manifestly obvious through a visual inspection that the risk to life from external fire spread is not such as to warrant a PAS 9980 assessment. This is particularly true in buildings with brick or masonry external walls or low risk buildings which do not present any significant risk of fire spread. In these cases, the fire risk assessor will normally address compliance of external wall construction with the Fire Safety Order as part of the routine fire risk assessment process.

A more detailed fire risk appraisal of the external walls under PAS 9980 may be required if, for example, there is a known or suspected risk from the form of construction used for the external wall, such as the presence of combustible materials used for cladding or external wall insulation.

Where a more detailed fire risk appraisal is required the Fire Risk Assessment Prioritisation Tool may be used to determine the urgency of the inspection.

Conducting a more detailed external walls assessment

In cases where a more detailed assessment is required, a competent person should be appointed. PAS 79-2:20202, PAS 9980 provide guidance, or if required EWS1 Form completed

17.4 The Building Safety Act 2022

This Act makes ground-breaking reforms to give residents and homeowners more rights, powers, and protections – so homes across the country are safer.

It delivers far-reaching protections for qualifying leaseholders from the costs associated with remediating historical building safety defects, and an ambitious toolkit of measures that will allow those responsible for building safety defects to be held to account.

It overhauls existing regulations, creating lasting change and makes clear how residential buildings should be constructed, maintained, and made safe.

The Act creates three new bodies to provide effective oversight of the new regime: the Building Safety Regulator, the National Regulator of Construction Products, and the New Homes Ombudsman.

Together these changes mean owners will manage their buildings better, and the home-building industry has the clear, proportionate framework it needs to deliver more, and better, high-quality homes.

Many of the detailed provisions in the Act will be implemented over the next two years through a programme of secondary legislation.

Protecting leaseholders

Through new leaseholder protection measures, the Act eradicates the idea that leaseholders should be the first port of call to pay for historical safety defects.

Building owners will not legally be able to charge qualifying leaseholders (defined under the Act but including those living in their own homes and with no more than three UK properties in total) for any costs in circumstances where a building (in the majority of cases meaning those over five storeys or eleven metres tall) requires cladding to be removed or remediated.

Qualifying leaseholders will also have robust protections from the costs associated with non-cladding defects, including interim measures like waking watches.

What the Act means for residents of higher risk buildings and homeowners

Residents in high-rise buildings will have more say in how their building is kept safe and will be able to raise building safety concerns directly to the owners and managers of their buildings known as accountable persons and responsible for repairing the common parts of a higher-risk building, as defined in section 72 of the Act. The accountable person(s) will have a duty to listen to them.

If residents feel their concerns are being ignored, they can raise them with the Building Safety Regulator.

All homeowners will also have more than twice the amount of time, from 6 to 15 years, to claim compensation for sub-standard construction work.

What the Act means for building owners

Duty holders such as the Principal Designer and Principal Contractor under the Act will be required to manage building safety risks, with clear lines of responsibility during the design, construction, and completion of all buildings.

Accountable persons will need to demonstrate that they have effective, proportionate measures in place to manage building safety risks in the higher-risk buildings for which they are responsible.

Those who do not meet their obligations may face criminal charges.

The Building Safety Act is also clear that building owners and landlords will need to contribute to the costs of fixing their own buildings.

What the Act means for the built environment industry

The Act will create a clear, proportionate framework for the design, construction, and management of safer, high-quality homes in the years to come.

It will strengthen the construction products regulatory regime, with new requirements to make sure all construction products on the UK market are safe for their intended use, with a National Regulator for Construction Products to monitor and enforce this.

There will be a new developer tax, and a levy on developers will ensure that industry contribute to setting things right.

New rights to redress will ensure those responsible for contributing to the building safety crisis are in line for costs to rectify their mistakes.

17.5 Construction, Design and Management Regulations

Projects undertaken in the UK are subject to the requirements of the Construction (Design and Management) Regulations 2015 (CDM), or within the European Union, that particular country's interpretation of the European Union Directive.

This report defines the strategy for meeting the functional and performance requirements for fire safety in the finished building. It is intended to form part of the submission for approval

under the Building Regulations, Part B (Fire safety). Where any conclusions or recommendations contained within this report specify particular materials, products, or forms of construction these will have been assessed, in accordance with CDM Regulation 9 (duties for designers).

In the event that these involve significant residual risks or health and safety critical assumptions, this information will be made available to the Principal Designer. Where the architect or other consultants use all or part of this report to specify works, they are understood to be competent in alerting the Client, Principal Designer, Designers, Contractors, and Building Occupier of issues arising under the CDM Regulations.

17.6 Statutory Consultation

During the Building Regulations application process, the Building Control body is required to formally consult with the local fire authority. The purpose of this consultation is to give the Fire Authority the opportunity to make observations in respect to the Building Regulations and to provide an opportunity to make the applicant aware of action that may have to be taken to meet the requirements of the Fire Safety Order.

The consultation should allow both parties to reach mutually compatible views on whether the building meets the requirements of both pieces of legislation. In the exceptional event that the Fire Authority propose to require physical changes to the building to meet the requirements of the Fire Safety Order, the Building Control body should make the applicant aware.